Detection and Forgetting under Asymmetric Data Availability in Class-Incremental Network Intrusion Detection

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Abstract

Class-incremental network intrusion detection must retain old attacks while learning new ones. Under explicit asymmetric data availability, AR-Stateless retrains a standard histogram-based gradient-boosted learner using refreshed benign data and attack-only memory. Five paired confirmation seeds yield final balanced accuracy of 65.4% on UNSW-NB15 and 98.3% on CIC-IDS-2017 at a 1% calibration false-positive-rate target, compared with ER-Bal-TabT's 56.3% and 92.7%. Signed forgetting is lower against this comparator in this view, but not uniformly across methods or decision views. On the previously exposed CSE-CIC-IDS2018 dataset, final balanced accuracy is 81.7%; the 2.1-point margin over the same-learner literature-memory control is narrow (paired one-sided t $p = 0.0491$, exact sign-flip $p = 0.0625$), and forgetting is higher than both that control and neural AR. Calibration targets are not guaranteed test false-positive rates. Historical neural-candidate failures remain part of the design-space evidence. All 60 added-baseline cells across three datasets are complete; the twelve original-rule low-FPR paired-t comparisons pass within-dataset Holm correction, but multiplicity-adjusted exact sign-flip sensitivity does not. A TEST-label-informed NSL-KDD diagnostic at identical measured 2% FPR instead favors neural AR over AR-Stateless (51.3% versus 47.4% BAcc), precluding a universal-superiority claim. Cross-learner and benign-access differences prevent attributing the comparisons to benign refresh alone. Standard learners, replay and calibration are not claimed as novel. These bounded prediction/retention results do not establish explanation stability, governance efficacy or completion of the substantive-innovation requirement. **Keywords:** continual learning; intrusion detection; benign data availability; false-positive calibration; replay memory.

1 Introduction

Intrusion detection models must recognize newly observed attack families without discarding knowledge of older ones. In class-incremental learning, historical attack records are represented by a limited replay memory, and inference must choose among all observed classes without a task identifier. This constraint is operationally meaningful when previous attack captures cannot be retained indefinitely or reacquired. It does not necessarily imply that benign traffic has the same availability: a monitored network may continue to provide benign examples.

The asymmetry is useful only under an explicit assumption. The learner must have access to a sufficiently clean, representative benign source, distinct from its test data. This paper does not assume that arbitrary production traffic is correctly labeled benign. Nor do we assume that a training-domain benign hold-out guarantees the same false-positive rate (FPR) after a distribution change. Both qualifications matter in our results.

Replay-based NIDS research includes class balancing, frequency-aware replacement, representative selection, and other memory designs [1–4]. Some multiclass formulations place Normal and attack classes in a common classifier and replay budget. However, this is not a universal description of the literature: binary and novelty-detection methods use different protocols, and prior domain-adaptation work has already used new benign data outside an old attack memory [5]. The question is therefore not whether benign data have ever been treated differently. This bounded study examines the detection, retention and cost trade-offs of stateless multiclass retraining with refreshed benign data and limited attack memory, under both default decisions and benign-only calibration. It does not identify benign refresh alone as the cause of a cross-method difference.

We study learning under asymmetric data availability. The selected AR-Stateless candidate fits a fresh multiclass tree learner at each task using newly sampled benign data, current attacks and retained old attacks. A disjoint benign-only calibration pool supplies the optional low-FPR operating point without test labels. The preceding AA-CIL design used a binary gate G and a separate attack-family classifier F; it and later neural retention variants are retained as historical design-space controls, not conflated with the selected stateless learner.

The contribution is an evidence-bounded experimental study:

- An explicit AR-Stateless data-access formulation separating persistent attack replay, refreshed benign training data, and permanent benign calibration data, with storage and training costs disclosed.
- Paired confirmation and published-method com-

parisons of the selected stateless learner, with completed dataset blocks distinguished from pending blocks and with backbone, access and adaptation differences retained.

- A joint default-decision and low-false-alarm evaluation reporting actual test FPR, binary attack detection, family-level performance, signed forgetting and negative registered outcomes rather than relying on aggregate accuracy alone.
- Post-hoc same-learner diagnostics of new-class exposure and memory size, with all-seen-data retraining identified as an offline reference rather than an admissible limited-memory competitor; these comparisons do not provide a complete causal decomposition.
- A retained historical neural design-space study, including modular gate/family controls, Replay-BS and neural AR, that records unsuccessful alternatives and the distinction between low forgetting and final predictive performance. This is comparative RQ2 evidence rather than a new-component or causal-mechanism claim; historical lead-reported summaries remain distinguished from independently generated results.

The historical neural design-space study follows the original superiority protocol, the validation-selected retention protocol, and the fixed-KD reassessment, followed by a fixed-capacity retention extension. Their criteria and chronology are given in Appendix D; each retains its own outcome.

2 Literature Review

2.1 Continual and class-incremental NIDS

Table 1 separates learning settings, benign-data roles, and reporting evidence among the verified papers we reviewed. The six-paper reporting sample includes binary domain-incremental learning and unsupervised novelty detection as well as multiclass Class-IL; it is neither a homogeneous benchmark nor an exhaustive survey.

CBRS-NIDS distinguishes balancing benign versus pooled attacks from balancing original classes [1]. ECBRS uses frequency-driven replacement and minority protection, whereas SSF selects and removes samples according to distributional relevance [2, 3]. These are predefined *rules*, not numerically fixed class quotas. SPIDER retains previous-task unlabeled samples and uses gradient-projection memory; CND-IDS instead regularizes latent representations and retains a clean Normal reference set [6, 7]. ACIL evaluates multiple replay strategies and explicitly reports forgetting [4]. ROC-AUC, benign PR-AUC, and Normal-class accuracy must not be substituted for a separately specified benign-FPR operating point.

The closest availability precedents prevent a universal memory-competition claim. Cerasuolo et al.'s Computer Networks model retains old benign and attack exemplars, with 25 exemplars per class rather than a demonstrated constant total budget [8]. Their EAAI study also receives target-domain benign data [9]. Most importantly, their RTSI adaptation adds benign data outside an old attack-only memory [5]. That experiment already establishes this broad availability pattern, which is therefore not claimed as novel here. AR-Stateless evaluates stateless multiclass retraining under repeated benign availability and a separate benign-only calibration role. TS-mem belongs to the historical gate/family study as a controlled restricted-refresh ablation, not a faithful reproduction of those systems or the selected stateless learner.

Wang and Cao describe anti-forgetting learning followed by unbiased prediction [10]. Only the official abstract and publication record were available: benign availability, replay contents, and a gate/family interpretation remain unverified. This paper is excluded from the reporting count, and missing full text is not novelty evidence.

Ariffin et al. [11] retain benign examples throughout replay, sharing our concern for continued benign exposure. Their single TabTransformer uses a shared buffer, without a separate gate or fixed-FPR calibration. They report accuracy, macro-F1, forgetting and intransigence on eight CICIDS2017 classes with seed 42, but no FPR. Their Section 5.7 reports replay final overall accuracy of 0.964/0.966/0.948 for MLP/CNN/FT-Transformer under a separate preprocessing pipeline; these are not balanced accuracies or matched comparisons with ours. Our registered backbone/capacity checks remain pending; no comparative advantage is established. This additional context does not change the six-paper sample below.

G1: Joint operating-point and retention reporting. Five of the six papers in the targeted reporting sample do not separately report fixed-operating-point benign FPR in their inspected main results; ACIL has partial or indirect benign reporting. The review record does not establish a joint forgetting-and-fixed-FPR detection evaluation in this sample. This bounded finding is not a claim that no prior paper reports such evidence, nor that unverified forgetting fields mean absence.

G2: Conditional competition for benign replay. Shared benign/attack memory in CBRS, ECBRS, SSF, ACIL, and the Computer Networks study motivates examining limited replay allocation. Competition under a fixed total budget is a setting-dependent constraint, not a universal property or an identified cause of forgetting. RTSI and CND-IDS are explicit counterexamples to a universal shared-buffer description. The selected stateless study measures prediction and retention under disclosed storage and exposure costs; isolated incremental effects of benign refresh are not established by comparisons that also change the learner. Separate family learning is a historical design-space factor, not

an AR-Stateless component.

2.2 Bias correction in Class-IL

BiC learns a post-training correction [12]; weight alignment adjusts classifier scale [13]; separated softmax and taskwise distillation address cross-task interference [14]; end-to-end incremental learning includes balanced fine-tuning [15]. Logit adjustment uses class-prior information to alter scores and is a long-tail classification approach, not by itself a Class-IL algorithm [16]. These are relevant controls for shared-classifier imbalance, but their distinct training, calibration, and inference operations must not be conflated.

G3: From class bias to benign-alert control. Correcting old/new-class bias does not by itself establish control of benign-to-attack false alarms or preservation of attack-family recall. In the selected AR-Stateless study, the relevant distinctions are the disclosed benign-data access and benign-only operating-point calibration of a multiclass learner; an independent binary gate belongs only to the historical neural study. These are protocol and endpoint distinctions, not proof that the original methods cannot be adapted, or that a binary gate is novel. Negative results for our tabular adaptations do not reject the original algorithms in general.

2.3 Replay and distillation

iCaRL combines exemplar retention and incremental representation learning [17]; Dark Experience Replay and DER++ retain historical responses alongside supervised rehearsal [18]. They supply replay and retention references, not ready-made evidence for decoupled NIDS. Their mechanisms and buffering schemes must not be treated as interchangeable. Image-domain results and original buffer policies are not automatically reproduced by tabular adaptations.

The historical fixed-capacity KD+WA candidate reuses distillation, DER-style replay, and weight alignment; none of these components is claimed as new. *The historical G3 evaluation question* was whether their combination in separate detection and family learners preserved old attacks while learning new ones at a disclosed false-alarm operating point. This is not the selected AR-Stateless architecture or a replacement for the locked research gap. That claim requires matched controls and joint accuracy/forgetting reporting, not simply the presence of KD or WA. The failed CSE confirmation reported below limits this candidate's claim rather than changing the literature gap. G1–G3 organize this bounded prediction/retention study under the broader RQ2; they do not replace the unanswered explanation-drift RQ1 or governance RQ3.

3 Problem Analysis

3.1 A corrected experimental basis

The motivating neural experiments use a taskwise minimum-update schedule rather than only a small fixed number of epochs; this optimizer-step rule does not describe the stateless tree learner. This prevents late, rare attack tasks from receiving arbitrarily few optimizer updates. Old attack classes are accessible only through stored training and validation rows after their introduction. Frozen preparation and duplicate/conflict handling are retained across methods; "deduplicated UNSW-NB15" does not imply that every benchmark is free of feature collisions or label conflicts. Historical experiments that revisited complete old attack pools are not used as evidence for the present access-constrained protocol. That old-pool resampling was a bug in this project's earlier implementation. The undertraining and data-quality issues identified in that development history are not attributed to published methods without source-specific evidence.

3.2 False alarms are important, but not a universal root cause

An earlier decomposition of the historical S2000 DER++ final confusion matrices showed substantial Normal-origin false positives in CIC-IDS-2017 attack-family columns, including families with high recall but low precision. These DER++ proportions must not be assigned to every method. UNSW-NB15 also exhibited attack-to-attack confusions and lost attack recall. These observations justify measuring benign FPR and attack detection separately, but do not identify inadequate benign coverage or Normal–attack memory competition as a universal causal explanation. The decomposition is arithmetic, not an intervention proving a mechanism.

Under the registered correction sweep, the best tested correction/base combinations closed only 4.92% and 19.23% of the positive Joint-minus-incremental Macro-F1 gaps on UNSW-NB15 and CIC-IDS-2017, respectively. This rejects the sufficiency of those specific adaptations and candidate ranges, not bias correction in general. Joint is an offline reference rather than a guaranteed numerical upper bound.

3.3 Why change the learning organization?

The preceding FARQ experiments adjusted shared replay allocation using measured validation false alarms. Their registered main criterion failed, and descriptive operating curves did not establish a universally superior false-alarm/detection frontier. Replay+LA and Replay share the same attainable predictions when the total adjustment is recalibrated; this does not make all allocation methods threshold-equivalent. WACB and Freq change the learned model and improve detection relative to Replay at evaluated matched-FPR points on UNSW-NB15 and CIC-IDS-2017. Memory reallocation

Table 1: Targeted NIDS literature comparison. The first six rows form the reporting sample. "Not separate" refers only to the inspected main results; "unverified" means the retained source record does not settle that field, not that the paper omits it. The last four rows supply availability context and are excluded from the six-paper count.

Work	Setting	Benign handling	Fixed-point benign FPR in main results	Forgetting evidence
CBRS-NIDS [1]	Binary output; sequential attacks	Shared replay; pooled-attack or original-class balancing	Not separate	Unverified
ECBRS/PAPA [2]	Binary domain-IL	Shared replay; frequency-driven replacement	Not separate	Unverified
SPIDER [6]	Semi-supervised binary domain-IL	Previous-task unlabeled samples; benign quota unverified	Not separate	Unverified
SSF [3]	Binary continual / drift	Shared fixed buffer; representative selection	Not separate	Unverified
CND-IDS [7]	Unsupervised binary novelty	Clean Normal reference set; no old-attack replay	Not separate	BWT reported; explicit forgetting unverified
ACIL [4]	Binary and multiclass	Shared replay; several strategies	Partial / indirect Normal reporting	Forgetting and BWT reported
Cerasuolo, ComNet [8]	Multiclass Class-IL	Normal in old-class replay; 25 exemplars per class	Unverified	Unverified
Cerasuolo, RTSI [5]	Multiclass; benign adaptation	New benign outside old attack-only memory; one adaptation	Unverified	Unverified
Cerasuolo, EAAI [9]	Cross-domain class adaptation	New-domain benign plus source-class memory	Unverified	Unverified
Wang and Cao [10]	Class-IL; details unverified	Abstract only, not verified	Unverified	Unverified

can therefore improve the measured trade-off, although it does not close every offline-reference gap. These finite operating-point comparisons are not a global frontier theorem. The historical AA-CIL study tested a different learning organization, not a premise that every shared-classifier method merely moves along one common curve.

4 Method: AR-Stateless under asymmetric data availability

4.1 Selected stateless learner

AR-Stateless (R3-A) uses one standard multiclass histogram-based gradient-boosted learner, including Normal, rather than the two neural networks described in Appendix A. Each task fits a new learner from scratch on the current new-class training rows, the retained old attack memory, and the task's benign training sample. No learned model parameters are carried into the next task. The attack-memory budget is $200K_t$, where K_t includes Normal; Normal's share is transferred to attacks. Old attack examples cannot be replenished from their original pools. The benign calibration pool is permanently excluded from fitting. Default inference chooses the highest scoring seen class; the low-FPR view separately calibrates the attack-versus- benign score. R3-L retains the same learner but stores 200 examples of each seen class, including Normal, without refreshed benign access.

Appendix A collects the historical neural gate/family architecture, update algorithm and retention variants. They are design-space controls, not the architecture or optimization procedure of AR-Stateless. Neural-specific update counts, cross-entropy, distillation and weight alignment do not apply to the tree. The shared data-access assumption is stated below; each historical round retains its own selection and outcome.

4.2 Data roles and access assumption

At update t, let $\mathcal{A}_t$ be the observed attack families and $\mathcal{N}_t$ the newly introduced attack families. The learner receives their current training data D_t^A, retained old attack training memory $R_{t-1}^{A,\mathrm{tr}}$, and a benign training source B^{tr}. A permanent benign calibration pool B^{cal} is disjoint from every optimizer input. We assume continued access to B^{tr} but not to the original pools of old attacks.

The benchmark simulates benign availability by drawing a task-dependent sample $B_t \subseteq B^{\mathrm{tr}}$ of up to 50000 rows. Samples may overlap across tasks; this is repeated access to a fixed prepared training source, not evidence from a newly collected live stream. The permanent split prevents a calibration row from later becoming a training row.

5 Experimental Setup

5.1 Benchmarks and task protocol

The historical neural studies use the prepared NSL-KDD, deduplicated UNSW-NB15, and CIC-IDS-2017 benchmarks [19–21]. The selected stateless confirmation and its external check have the separately stated dataset scope below. Normal and an initial attack family appear in the first task, followed by the fixed attack-family sequence. The data manifest and generated protocol table specify label mappings, task groupings, split sizes, and feature dimensions. Test rows from unobserved classes are excluded at intermediate checkpoints; final metrics use the complete prepared test set. No test row is selected for training, memory construction, or primary calibration. The additional withheld CSE-CIC-IDS2018 assessment uses the bound full cache with 12,882,081 training rows, 3,255,102 test rows and 78 features. Its four tasks are: Normal, DoS; DDoS, Bruteforce; WebAttack, Infiltration; Botnet. The same full-cache rows, task order, update rule and calibration policy are bound across methods. The permanent Normal calibration holdout contains 1,070,273 rows (342,487,360 feature/index bytes), outside replay memory and never optimized. These fields are taken from the released execution identities, not inferred from the dataset name.

5.2 AR-Stateless confirmation under asymmetric availability

The later R3-A candidate (AR-Stateless) retrains a standard histogram-based gradient-boosted learner at each task using fresh benign data and attack-only memory. Its total memory budget is 200 times the number of seen classes, including the benign share, distributed across attacks. R3-L uses the same learner with 200 stored examples per seen class including benign, without a refreshed benign stream. This tests the asymmetric-availability design, not a novel tree algorithm; both designs remain bounded RQ2 studies.

Tables 4 and 5 reuse completed seeds 10–14. At the 1% calibration target, R3-A improves final balanced accuracy over ER-Bal-TabT by 9.14 percentage points on UNSW and 5.53 on CIC-2017, with lower signed forgetting in this view. This does not imply lower forgetting in every view or against every comparator. On CSE-2018, gains over AR and R3-L are 2.63 and 2.06 points. The latter is narrow: one-sided paired t $p = 0.04914$, exact sign-flip $p = 0.0625$. R3-A's signed forgetting is 7.3 points, higher than AR's -3.5 and R3-L's 3.1. New seeds on the previously exposed CSE dataset are not independent unseen-dataset replication. Historical candidate failures remain unchanged. These results do not establish explanation continuity or governance efficacy, or close the substantive-innovation gate. Added published baselines and their separately registered S1 comparison are now complete on all three datasets, as reported below; this does not close the broader scientific requirements.

5.3 Historical neural training and hardware

The earlier MLP-based neural experiments use two hidden layers of width 128, Adam with learning rate 0.001, and batch size 256. Each model/update receives

$$S_t = \max(2000, 8\,\lceil |D_t|/256 \rceil) \tag{1}$$

optimizer steps. Consequently, matching this schedule does not imply identical realized steps when data exposure differs. Gate and family schedules are recorded separately. The registered seed set contains 5 seeds; tables report actual completed n where the cutoff prevents a complete group. Partial groups are not silently replaced with selected favorable seeds.

The historical neural execution records include an NVIDIA GeForce RTX 4060 Ti with 8 GiB memory locally and NVIDIA A100 GPUs on HPC. These records do not describe the CPU execution of AR-Stateless or imply a current HPC allocation. HPC results are included only after completed execution, not merely submission or allocation. Heterogeneous hardware and shared local GPU use mean that reported runtimes are not a controlled hardware benchmark.

5.4 Historical neural comparators and ablations

The original neural external comparison set is Replay, DER++, WACB, Freq, and Replay+LA. The later stateless and added-baseline comparisons retain their own comparator sets and adaptation identities. Replay uses the common tabular architecture. DER++ is the recorded tabular adaptation with classwise memory management, not an exact replication of the original vision protocol. WACB adapts benign-versus-combined-attack balancing to this multiclass setting. Freq is a simplified frequency-proportional allocation control, not an exact ECBRS reproduction. Replay+LA is a saved-model logit-adjustment evaluation; it is not an additional independently trained network. All primary operating thresholds are determined from the same permanent benign calibration pool.

TS-mem retains the same gate/family organization but restricts each task's benign refresh to the training portion of the ordinary Normal allocation, while sharing replay capacity between Normal and attacks. It therefore measures a restricted benign exposure policy, not the effect of a frozen benign subset alone. AA-noAttMem retains refreshed benign training and the independent gate but does not transfer the Normal allocation to attacks. Its name does not mean that attacks have no replay memory.

Joint trains with the offline data available under its recorded protocol and is reported as a reference. It is not a guaranteed upper bound, a matched-storage online learner, or a member of the strongest-baseline selection. FARQ is supplementary. The registered selection includes the external methods *and* TS-mem and AA-noAttMem; excluding those ablations after inspecting results would change the registered question.

Table 2: Dataset/task and final-budget summary. Calibration bytes include float32 features and int64 row indices, exclude model state and dense labels. The pool is shared by all methods, excluded from optimization, and outside M.

Dataset	Classes	Tasks	M ceiling	Calib rows	Calib bytes
NSL-KDD	5	4	1000	6735	3178920
UNSW-NB15	10	5	2000	6825	1911000
CIC-IDS-2017	8	4	1600	181176	57251616

Table 3: Prepared split sizes, feature counts and fixed task groupings (class IDs). These are the exact prepared benchmark arrays, not an assertion of temporal deployment validity.

Dataset	TRAIN rows	TEST rows	Features	Task IDs (semicolon separates tasks)
NSL-KDD	125973	22544	116	0,1; 2; 3; 4
UNSW-NB15	120991	30248	68	0,1; 2,3; 4,5; 6,7; 8,9
CIC-IDS-2017	2262203	565673	77	0,1; 2,3; 4,5; 6,7

NSL-KDD: 0=Normal, 1=DoS, 2=Probe, 3=R2L, 4=U2R. UNSW-NB15: 0=Normal, 1=Generic, 2=Exploits, 3=Fuzzers, 4=DoS, 5=Reconnaissance, 6=Analysis, 7=Backdoor, 8=Shellcode, 9=Worms. CIC-IDS-2017: 0=Normal, 1=DoS, 2=DDoS, 3=Bruteforce, 4=PortScan, 5=WebAttack, 6=Botnet, 7=Infiltration.

5.5 Metrics and registered inference

For confusion counts C_{ij} with Normal index b, we report

$$\mathrm{FPR} = \frac{\sum_{j \neq b} C_{bj}}{\sum_j C_{bj}}, \tag{2}$$

$$\mathrm{ADR} = \frac{\sum_{i \neq b} \sum_{j \neq b} C_{ij}}{\sum_{i \neq b} \sum_j C_{ij}}. \tag{3}$$

ADR is binary attack detection, not correct-family recall. Macro-F1 and attack-family macro recall supplement it. The registered family-recall guard includes attack classes meeting the specified minimum test support; excluded rare classes remain visible in per-class evidence. Precision is undefined for a class with no predictions in a seed and is never changed to zero. Its group mean and sample standard deviation use only defined seed values, with the actual n_{defined} (`defined_n`) reported separately from the total seed count. If none are defined, the mean remains undefined; with fewer than two defined values, the sample standard deviation is undefined. Such a seed is not dropped from recall, forgetting, or other defined endpoints.

The original neural protocol's primary calibration target is $\alpha = 2\%$; the later stateless confirmation uses the separately registered low-FPR target reported above. Actual test FPR is reported alongside it because equal calibration targets need not yield equal test FPR. The original superiority protocol requires a one-sided paired test against the highest-mean registered comparator in at least two datasets. The comparator is chosen once per dataset and endpoint from whole-seed means, not separately per seed. Other criteria concern Macro-F1 harm, attack-family recall, the CIC gate-refresh ablation, and descriptive cost/Joint reporting. Paired differences, sample standard deviations and effect sizes accompany complete five-seed comparisons. Appendix D defines the historical protocols and their distinct comparator sets.

5.6 Why a fixed false-alarm budget

False alarms affect the usefulness of detection when attacks are rare: the base rate enters alarm precision, so high detection alone is insufficient [22]. Operational errors also have substantial costs [23]. Arp et al.'s P7 cautions against unsuitable performance measures; their P8 recommends precision/recall analysis under realistic prevalence, not an exclusive fixed-FPR metric [24]. We therefore retain precision, recall and the default-decision view alongside budgeted evaluation. Published precedents include EMBER's malware detection rates below 1% and 0.1% FPR [25], and Kitsune's network-anomaly TPR/FNR at FPR 0 and 0.001 [26]. Neither establishes 1% as a universal class-incremental NIDS standard.

Our empirical motivation is that default decisions incur different alarm burdens: final UNSW FPR averages 25.07% for R3-L and 1.26% for AR-Stateless over seeds 10–14. Their default accuracies remain valid measurements of their own decision policies, but do not compare detection at a matched false-alarm budget. After each task calibration uses the permanent, optimization-excluded benign pool and a common empirical false-alarm cap. The decision maps are not identical: AR-Stateless and neural gates use a strict score threshold; the multiclass Replay-BS/Freq/WACB path selects a feasible logit-prior adjustment by bracketing and bisection. The historical permanent pool was also available to allocation validation, so it is not a fresh independent calibration sample. This is a Neyman–Pearson-inspired operating constraint, not an implementation of Tong et al.'s umbrella algorithm or a finite-sample population-error guarantee [27]. Indeed, that work distinguishes empirical type-I error control from high-probability population control. Across all stages and seeds 10–14, AR-Stateless's measured test FPR at the 1% target ranges from 0.75–1.15% on UNSW, 0.94–1.06% on CIC-2017, and 0.88–1.04% on CSE-2018. Historical Section 46 NSL-KDD external-baseline records (Replay, Freq, WACB, DER++ and Joint; seeds 1–4/42; four stages each) span 0.75–7.76% at the 1% target, independently recomputed from 100 confusion ma-

Table 4: Confirmation seeds 10–14: mean ± sample standard deviation. All metrics use the 1% calibration FPR target (not a guaranteed test FPR). BAcc is macro recall over seen classes; incremental BAcc averages stages. Signed forgetting averages best pre-final minus final recall over old classes, including Normal; negative values are retained. Values are percentages or percentage points.

Dataset	Method	Incremental BAcc	Final BAcc	Signed forgetting
UNSW	R3-A	74.4 ± 0.4	65.4 ± 0.7	20.6 ± 0.5
UNSW	R3-L	61.2 ± 1.4	45.8 ± 2.3	20.9 ± 0.9
UNSW	ar	66.3 ± 0.8	51.0 ± 1.6	9.5 ± 1.3
UNSW	replay_bs	59.3 ± 0.7	33.2 ± 1.9	5.0 ± 1.8
UNSW	er_bal_tabt	69.8 ± 0.5	56.3 ± 1.6	25.0 ± 1.0
UNSW	freq	52.5 ± 1.6	27.5 ± 1.2	2.4 ± 1.7
UNSW	wacb	50.5 ± 3.0	27.2 ± 2.4	1.4 ± 2.5
CIC-2017	R3-A	99.1 ± 0.2	98.3 ± 1.1	0.4 ± 0.1
CIC-2017	R3-L	97.0 ± 0.9	92.9 ± 1.9	2.5 ± 1.4
CIC-2017	ar	86.3 ± 1.1	81.6 ± 3.0	-3.1 ± 1.5
CIC-2017	replay_bs	80.3 ± 2.4	57.4 ± 6.0	11.4 ± 5.1
CIC-2017	er_bal_tabt	96.7 ± 0.6	92.7 ± 2.0	2.7 ± 0.3
CIC-2017	freq	69.3 ± 3.1	48.1 ± 4.7	8.5 ± 5.6
CIC-2017	wacb	68.4 ± 1.5	45.9 ± 2.2	5.1 ± 4.3
CSE-2018	R3-A	88.7 ± 0.8	81.7 ± 1.0	7.3 ± 2.7
CSE-2018	R3-L	83.8 ± 0.7	79.7 ± 2.1	3.1 ± 3.6
CSE-2018	ar	83.2 ± 3.6	79.1 ± 2.2	-3.5 ± 8.4
CSE-2018	replay_bs	82.9 ± 3.8	68.1 ± 13.9	11.9 ± 7.9

Table 5: Paired one-sided t tests and exact sign-flip tests (32 assignments). Differences are first minus second method. R3 tests whether R3-A forgetting is higher than ER; its registered rule requires t p $\geq$ .05. Failure to detect harm is not equivalence or non-inferiority. Other criteria require superiority with t p $<$ .05. Tests are unadjusted as registered for R1–R5; the separate added-baseline S1 Holm family is not evaluated here.

Dataset	Comparison	Difference (pp)	t p	Sign-flip p
UNSW	R1 final: R3-A > ER	9.136	2.67577e-05	0.03125
UNSW	R2 incremental: R3-A > ER	4.608	9.94844e-07	0.03125
UNSW	R3 harm: R3-A > ER	-4.422	0.998722	1.00000
UNSW	R4 final: R3-A > R3-L	19.611	8.77594e-06	0.03125
CIC-2017	R1 final: R3-A > ER	5.525	0.000818287	0.03125
CIC-2017	R2 incremental: R3-A > ER	2.446	0.000276361	0.03125
CIC-2017	R3 harm: R3-A > ER	-2.298	0.999924	1.00000
CIC-2017	R4 final: R3-A > R3-L	5.309	0.00123769	0.03125
CSE-2018	R5a final: R3-A > AR	2.632	0.0124407	0.03125
CSE-2018	R5b final: R3-A > R3-L	2.058	0.0491421	0.06250

trices. This demonstrates calibration shift, not a universal 3–8% range or a result for the new confirmation seeds.

The UNSW pool contains 6,825 benign rows and CIC-2017 contains 181,176. Under a strict empirical $\leq \alpha$ cap, UNSW permits at most $\lfloor 6825\alpha \rfloor$ false alarms: 68 at 1%, but only 6 at 0.1% (not 7). This illustrates resolution and tail-sample limitations, not operational acceptability of 1%. The 2% and 5% points provide sensitivity checks. Historically, work-order Section 27 used 2% as primary; Section 48 moved the low-FPR claim to 1%; Section 165 registered R1–R5 before the seed-10–14 confirmation runs. This sequence does not make the earlier development evidence unexposed. ER-Bal-TabT (official-code-based adaptation), Freq, WACB, Replay-BS, R3-L and neural AR are evaluated under the common budget protocol while retaining their stated data-access rules. The Section 178 baselines join this comparison only when completed and verified. Budgeted evaluation is an evaluation choice, not a proposed mechanism or contribution.

6 Main results: default decisions

The historical confirmation-source comparison uses uncalibrated decisions: argmax over seen classes for single classifiers, and a strict gate probability $P(\text{attack}) > 0.5$ followed by the seen-family argmax for two-stage methods. No calibration data or FPR threshold search enters this view. Tables 6–8 report final accuracy, macro-F1 and weighted-F1, mean stage accuracy (ACC), old-class recall BWT, and signed forgetting. BWT compares final recall with recall at class introduction; forgetting compares it with the best preceding recall. Both include Normal and weight old classes equally. These are different estimands from stage balanced accuracy and from calibrated retention below.

These tables retain every requested comparator by name and show default-view seed blocks absent from the historical confirmation/default-export collections used by `build_default_view.py`. They do not read the later added-baseline collection: in particular, the DER++ NA entries here do not mean that the completed DER++ blocks in Table 10 are unavailable. The original numerical rows and exploratory comparisons are retained without pooling the later method family into this snapshot. The fresh seed block is 10–14; historical seeds are not mixed into it to fill missing cells. ER default exports are now complete on both datasets. No calibrated confusion matrix is relabelled as a default decision. Earlier results and their actual seed counts remain reported in the subsequent analysis and appendices.

At default decisions, historical neural AR exceeds

Table 6: UNSW-NB15: historical confirmation-source snapshot (default decision, no FPR calibration). Seeds 10–14, mean ± sample SD; percentages or percentage points. ACC is mean stage micro accuracy; BWT is mean old-class recall change since introduction; forgetting is signed best pre-final minus final old-class recall, including Normal. NA denotes absence from these historical confirmation/default-export collections, not a zero or an omitted failed seed. Later completed added-baseline blocks are reported separately in Table 10.

Method	$n/5$	Accuracy	Macro-F1	Weighted-F1	ACC	BWT	Forgetting
AR-Stateless	5	76.9 ± 0.3	51.2 ± 0.8	73.7 ± 0.4	83.1 ± 0.1	-27.4 ± 0.6	27.4 ± 0.6
R3-L	5	72.7 ± 0.8	50.8 ± 0.7	76.4 ± 0.7	78.7 ± 0.3	-21.1 ± 1.0	21.1 ± 1.0
Replay	0	NA	NA	NA	NA	NA	NA
DER++	0	NA	NA	NA	NA	NA	NA
WACB	5	68.9 ± 0.9	40.3 ± 1.0	72.0 ± 0.9	77.4 ± 0.5	-29.6 ± 0.9	29.6 ± 0.9
Freq	5	71.2 ± 0.8	41.7 ± 0.9	73.9 ± 0.6	78.4 ± 0.3	-35.9 ± 1.3	35.9 ± 1.3
Replay-BS	5	69.7 ± 1.4	43.0 ± 0.9	73.9 ± 0.8	78.6 ± 0.3	-22.3 ± 2.1	22.4 ± 2.1
AA-CIL	0	NA	NA	NA	NA	NA	NA
KD+WA	0	NA	NA	NA	NA	NA	NA
AKD	0	NA	NA	NA	NA	NA	NA
Joint (offline)	0	NA	NA	NA	NA	NA	NA
AR	5	74.6 ± 0.7	45.3 ± 0.7	75.2 ± 0.3	82.6 ± 0.2	-10.9 ± 1.8	13.1 ± 1.3
AR-Joint	5	68.5 ± 1.5	36.6 ± 1.6	70.2 ± 0.9	78.4 ± 1.3	-36.7 ± 3.3	36.7 ± 3.3
ER-Bal-TabT	5	73.0 ± 0.3	42.6 ± 0.1	69.4 ± 0.4	81.2 ± 0.1	-27.1 ± 2.4	27.1 ± 2.4

ER-Bal-TabT on UNSW by 1.63 percentage points in final accuracy and 2.74 points in macro-F1, whereas on CIC it trails by 3.36 and 14.77 points, respectively. These are dataset-dependent descriptive differences, distinct from calibrated balanced accuracy. The two-sided paired t tests and exact sign-flip tests are shown together, without multiplicity adjustment; with five pairs the smallest two-sided exact sign-flip p is 0.0625. Thus a small paired-t value alone is not treated as robust confirmation across tests. These exploratory comparisons do not revise K1 or any historical decision.

6.1 Comparison with literature conventions

The default view supplies conventional multiclass metrics under this study's prepared data, task order and benign-access assumption. The separate supervisor-group protocol check uses CICIDS2017's eight classes, four experiences, 60/20/20 split, CI and CII settings, 10% memory, and seeds 42, 1, 2, 3, 4. Its different protocol must not be pooled with these default-view tables or described as completed parity evidence without its own results. Under our low-FPR protocol, ER-Bal-TabT remains more accurate than neural AR on both UNSW and CIC; this is not a claim about all deployments or a faithful replication of every supervisor-group setting.

6.2 Additional published-method adaptations

Tables 10 and 11 add the completed seed-10–14 blocks to the AR-Stateless reference. Every displayed value is regenerated from published confusion counts by `export_published_baselines.py`; no historical seed fills a missing block. Table 12 now records all 60 primary cells completed and published across three datasets and four adaptations, with five seeds per block. Five UNSW author-memory and five historical native-schedule BiC sensitivity cells remain additional and excluded from primary coverage and Holm families. The original prior-shift registered S1 comparisons pass within each dataset, including the completed CSE-2018 block. These tests concern the specified calibrated endpoint only; they do not establish homogeneous historical runtimes, universal superiority, explanation-drift or governance efficacy, or completion of the substantive-innovation requirement.

The default-view advantage of DER++ on UNSW must remain visible: its final BAcc is 68.46%, above AR-Stateless's 65.99%, but at 24.05% versus 1.26% FPR. DER++ has lower accuracy (68.42% versus 76.91%) and macro-F1 (43.67% versus 51.17%). At the 1% calibration target its final BAcc is 12.17%; the default ranking is not a matched-alarm-budget comparison. On CIC-2017, DER++ changes from 92.35% BAcc at 10.87% FPR to 35.75% BAcc at 0.99% measured test FPR. The iCaRL adaptation changes from 73.77% BAcc at 47.69% FPR to 26.47% BAcc at 0.94% FPR. These large operating-point trade-offs are not evidence of universal method superiority.

The neural implementations are tabular adaptations. DER++ retains the disclosed classwise memory and task-boundary schedule, rather than reproducing the original online vision protocol. iCaRL uses herding, softmax-KL distillation and nearest-mean-of-exemplars inference, not the original sigmoid-BCE objective. Their stored feature memories allow membership checks against permitted training pools, but do not retain unique originating row IDs; repeated feature vectors cannot be treated as proof of distinct source samples. For the five CIC-2017 cells, DER++ checkpoint training totals 1622.24 seconds. iCaRL totals 1167.21 seconds of checkpoint training and an additional 157.21 seconds of buffer selection; its end-to-end summaries total 1406.25 seconds. Sampled peak resident memory is about 2.23 and 3.01 GB, respectively. These local CPU measurements include separate selection costs and are not hardware-independent efficiency rankings.

ECBRS is adapted to the same multiclass learner and task protocol, not its native binary training entry point. The primary paper-memory implementation uses $M\,\mathrm{softmax}(-n_c)$ and global encounter-frequency ranking for eligible victims; this interpretation is supported by the supplementary worked example, while the main algorithm's local-maximum wording leaves a

Table 7: UNSW-NB15: exploratory paired default-decision comparisons within the historical source collections, historical neural AR minus comparator, seeds 10–14. Two-sided paired t and exact sign-flip p values (all 32 assignments), unadjusted for multiple comparisons. Only complete paired blocks are tested; these tests do not replace historical decisions. Positive accuracy/F1/ACC/BWT differences favor AR; negative forgetting differences favor AR.

Comparator	Endpoint	Mean difference (pp)	p_t	p_{flip}
AR-Stateless	Accuracy	-2.30	0.00113	0.0625
AR-Stateless	Macro-F1	-5.85	0.00017	0.0625
AR-Stateless	Weighted-F1	1.49	0.00129	0.0625
AR-Stateless	ACC	-0.46	0.00556	0.0625
AR-Stateless	BWT	16.48	1.97e-05	0.0625
AR-Stateless	Forgetting	-14.30	7.32e-06	0.0625
R3-L	Accuracy	1.90	0.0114	0.0625
R3-L	Macro-F1	-5.52	0.000107	0.0625
R3-L	Weighted-F1	-1.17	0.00289	0.0625
R3-L	ACC	3.91	2.83e-05	0.0625
R3-L	BWT	10.11	0.000817	0.0625
R3-L	Forgetting	-7.93	0.000975	0.0625
WACB	Accuracy	5.70	0.000352	0.0625
WACB	Macro-F1	4.97	0.000529	0.0625
WACB	Weighted-F1	3.17	0.00263	0.0625
WACB	ACC	5.22	2.92e-06	0.0625
WACB	BWT	18.65	1.08e-05	0.0625
WACB	Forgetting	-16.48	9.36e-06	0.0625
Freq	Accuracy	3.44	0.00132	0.0625
Freq	Macro-F1	3.62	0.00162	0.0625
Freq	Weighted-F1	1.29	0.0276	0.0625
Freq	ACC	4.23	2.9e-06	0.0625
Freq	BWT	25.00	2.02e-05	0.0625
Freq	Forgetting	-22.83	8.4e-06	0.0625
Replay-BS	Accuracy	4.88	0.00391	0.0625
Replay-BS	Macro-F1	2.28	0.0203	0.1250
Replay-BS	Weighted-F1	1.35	0.0571	0.1250
Replay-BS	ACC	4.00	1.09e-05	0.0625
Replay-BS	BWT	11.35	0.00174	0.0625
Replay-BS	Forgetting	-9.26	0.00178	0.0625
AR-Joint	Accuracy	6.14	0.00236	0.0625
AR-Joint	Macro-F1	8.72	0.00111	0.0625
AR-Joint	Weighted-F1	5.05	0.000459	0.0625
AR-Joint	ACC	4.19	0.00289	0.0625
AR-Joint	BWT	25.72	0.000106	0.0625
AR-Joint	Forgetting	-23.55	0.000152	0.0625
ER-Bal-TabT	Accuracy	1.63	0.00246	0.0625
ER-Bal-TabT	Macro-F1	2.74	0.00149	0.0625
ER-Bal-TabT	Weighted-F1	5.82	1.41e-06	0.0625
ER-Bal-TabT	ACC	1.38	4.95e-05	0.0625
ER-Bal-TabT	BWT	16.21	0.000268	0.0625
ER-Bal-TabT	Forgetting	-14.03	0.000564	0.0625

semantic ambiguity. It is not claimed to be the uniquely faithful interpretation. Under the implemented pre-decrement eligibility test, a positive allocation below one can permit removal of a class's final slot even without floating-point underflow. The observed UNSW block has 98.63% default FPR and 54.35% BAcc; these poor results are retained. In seed 10, Normal memory is empty from the second task onward. This software-level mechanism check does not identify the cause of the entire performance loss or show that the native binary algorithm must collapse.

The separate author-memory sensitivity ports the fixed author implementation's replacement and retrieval rules, with an explicitly documented underfill-index repair, row-ID payload adaptation, growing capacity and no native first-task prefill/skip. Its UNSW default BAcc/FPR are 66.20%/38.49%, and its 1%-calibrated BAcc is 14.38%. Several memory rules change together; the difference cannot be attributed solely to γ or retained Normal slots. Both versions and their full execution evidence remain available. The B-CER comparison selects BiC because Table 6 of Cerasuolo et al. [8] reports the lowest all-class macro-F1 Drop among its eight CIL methods on all four datasets: 10.92%, 5.48%, 18.16%, and 12.85% for IoT-23, TON-IoT, IoT-NID, and Edge-IIoT, respectively; these are source-paper values, not results on our registered splits.

B-CER is explicitly a flow-vector Conv1D adaptation, not a reproduction of the native packet-level 10×4 input. It preserves the registered feature order and splits, two convolution/pooling/normalization stages, a dense layer, expanding outputs, distillation, and a separate frozen-backbone bias-correction stage. The channel counts 32/64, kernel size 3, pool size 2, and dense width 128 are engineering defaults, not claimed paper parameters. The native backbone schedule is at most 200 epochs, patience 20, initial learning rate 0.1 divided by 3 down to 10^{-4}. The first CIC-2017 native-schedule cell reached the 55-minute local time budget (3304.72 s monitored wall time) before completing its first task; this is incomplete execution, not an algorithm-performance result. Accordingly, the entire CER primary family uses a disclosed common stage-1 override: Adam at learning rate 0.001, batch size 256, at least eight full epochs and 2000 optimizer steps, no early stopping or learning-rate decay, and the final iterate. The TRAIN stopping subset supplies diagnostic loss only and does not select the model. Stage 2

Table 8: CIC-IDS-2017: historical confirmation-source snapshot (default decision, no FPR calibration). Seeds 10–14, mean ± sample SD; percentages or percentage points. ACC is mean stage micro accuracy; BWT is mean old-class recall change since introduction; forgetting is signed best pre-final minus final old-class recall, including Normal. NA denotes absence from these historical confirmation/default-export collections, not a zero or an omitted failed seed. Later completed added-baseline blocks are reported separately in Table 10.

Method	$n/5$	Accuracy	Macro-F1	Weighted-F1	ACC	BWT	Forgetting
AR-Stateless	5	99.4 ± 0.1	89.6 ± 2.0	99.4 ± 0.1	99.6 ± 0.0	-1.1 ± 0.2	1.1 ± 0.2
R3-L	5	95.1 ± 1.2	62.1 ± 3.0	96.5 ± 0.8	97.1 ± 0.4	-1.3 ± 0.3	1.3 ± 0.3
Replay	0	NA	NA	NA	NA	NA	NA
DER++	0	NA	NA	NA	NA	NA	NA
WACB	5	91.8 ± 1.2	55.3 ± 1.6	93.7 ± 0.9	93.9 ± 0.6	-3.5 ± 0.8	3.5 ± 0.8
Freq	5	92.4 ± 1.1	56.7 ± 1.1	94.0 ± 0.9	94.4 ± 0.7	-6.3 ± 1.0	6.3 ± 1.0
Replay-BS	5	91.8 ± 1.1	57.3 ± 1.5	93.7 ± 0.7	95.9 ± 0.3	-3.4 ± 1.2	3.4 ± 1.2
AA-CIL	0	NA	NA	NA	NA	NA	NA
KD+WA	0	NA	NA	NA	NA	NA	NA
AKD	0	NA	NA	NA	NA	NA	NA
Joint (offline)	0	NA	NA	NA	NA	NA	NA
AR	5	95.2 ± 0.4	62.8 ± 0.6	96.1 ± 0.3	96.9 ± 0.3	-4.2 ± 1.5	4.4 ± 1.4
AR-Joint	5	95.3 ± 0.2	63.4 ± 0.6	96.1 ± 0.2	97.0 ± 0.2	-5.0 ± 0.8	5.2 ± 0.8
ER-Bal-TabT	5	98.6 ± 0.2	77.6 ± 5.2	98.6 ± 0.2	98.9 ± 0.1	-8.3 ± 7.8	8.3 ± 7.8

retains the separately disclosed engineering defaults of 200 SGD epochs, learning rate 0.1, momentum 0.9, and a 0.05 squared-bias penalty; its cost is reported separately. Our budget is up to 200 unique rows per seen class, including Normal and the correction subset, with no old-class refresh, rather than the paper's 25 exemplars per class. All five cells on each of UNSW, CIC-2017 and CSE-2018 use this common configured identity; the CSE checkpoint-resume and runtime limits are disclosed below. The original five native-schedule UNSW cells remain a separate historical sensitivity, excluded from primary coverage and Holm comparisons; no schedules are mixed within the primary family. Software smoke validation does not enter the scientific comparison.

The historical native-schedule UNSW BiC adaptation has default BAcc 68.50 ± 1.11%, accuracy 67.32%, macro-F1 44.04%, FPR 28.53%, and signed forgetting 19.50 pp. At the 1% calibration target, BAcc falls to 17.28 ± 2.75%, with test FPR 0.84%, attack recall 33.11%, and forgetting 12.04 pp. Thus its default recognition comes with high false-alarm exposure, while low-FPR recognition remains weak; these descriptive results do not establish S1 superiority. Across the five seeds, stage-1 training takes 6933.52 s and bias fitting 2.53 s (6936.04 s combined); herding is separately measured at 13.54 s. Summary elapsed time is 7074.61 s, monitored wall time 7115.61 s, and sampled peak child RSS 0.680 GB (decimal). These overlapping elapsed measures must not be added as independent costs. Full checkpoint hashes, retained row IDs, native-stage-1 schedule evidence, and correction-stage engineering defaults remain attached to the published historical group.

Under the common stage-1 override, UNSW BiC has default BAcc 66.71 ± 1.38%, accuracy 65.93%, macro-F1 43.28%, FPR 28.45%, and signed forgetting 18.48 pp. At the 1% calibration target, BAcc is 15.44±2.38%, test FPR 1.03%, attack recall 21.02%, and forgetting 13.97 pp. The high default false-alarm rate and weak low-FPR recognition remain explicit; no superiority or causal attribution to one optimizer setting is claimed. Across five seeds, stage-1 training takes 972.71 s, bias fitting 2.35 s (975.06 s combined), and herding 12.38 s. Summary elapsed time is 1063.35 s, monitored wall time 1098.00 s, and sampled peak child RSS 0.690 GB (decimal); overlapping elapsed measures are not additive. Source-bound checkpoints, real retained row IDs, all decision views, and the native CIC-2017 time-budget failure evidence accompany the new common-budget group.

The completed common-budget CIC-2017 BiC block has default BAcc 88.32 ± 6.24%, accuracy 83.31%, macro-F1 48.69%, FPR 18.72%, and signed forgetting 7.92 pp. At the 1% calibration target, BAcc is 35.87 ± 3.31%, test FPR 0.98%, attack recall 59.64%, and forgetting 14.85 pp. At 2% and 5%, BAcc/FPR are 44.41/1.99% and 65.67/4.88%, respectively. The default false-alarm burden, weak low-FPR recognition, and seed variability are retained; these descriptive results do not establish S1 superiority or native packet reproduction. Across five seeds, stage-1 training takes 5369.51 s, bias fitting 1.76 s (5371.28 s combined), and herding 369.38 s. Summary elapsed time is 6352.63 s, monitored wall time 6397.98 s, and sampled peak child RSS 2.451 GB (decimal); these overlapping measures are not additive. All four decision views, real retained row IDs, checkpoint/progress hashes, and the unchanged bound sources accompany the published group.

The completed CIC-2017 ECBRS paper-memory block yields 70.79% default BAcc but 81.09% FPR, 25.55% accuracy, and 31.86 pp signed forgetting. At the 1% calibration target, BAcc is 19.11% at 0.98% measured test FPR, with 14.52 pp forgetting. The same implementation limits apply; these results do not establish a failure of the native binary method. Across five seeds, checkpoint training totals 3151.76 s and summary elapsed time totals 3200.69 s; sampled peak RSS is 2.32 GB (decimal). Final memories each contain 1600 slots but only 1137, 1127, 1131, 1132, and 1137 distinct source row IDs, respectively. Unlike the feature-only diagnostics above, these are row-ID counts; repeated presentations can occupy multiple slots.

The completed CSE-2018 DER++ block has default BAcc 81.65 ± 2.06%, accuracy 73.19%, macro-F1

Table 9: CIC-IDS-2017: exploratory paired default-decision comparisons within the historical source collections, historical neural AR minus comparator, seeds 10–14. Two-sided paired t and exact sign-flip p values (all 32 assignments), unadjusted for multiple comparisons. Only complete paired blocks are tested; these tests do not replace historical decisions. Positive accuracy/F1/ACC/BWT differences favor AR; negative forgetting differences favor AR.

Comparator	Endpoint	Mean difference (pp)	p_t	p_{flip}
AR-Stateless	Accuracy	-4.21	2.92e-05	0.0625
AR-Stateless	Macro-F1	-26.76	4.84e-06	0.0625
AR-Stateless	Weighted-F1	-3.32	1.88e-05	0.0625
AR-Stateless	ACC	-2.67	1.61e-05	0.0625
AR-Stateless	BWT	-3.17	0.00819	0.0625
AR-Stateless	Forgetting	3.31	0.00603	0.0625
R3-L	Accuracy	0.08	0.89	0.8750
R3-L	Macro-F1	0.72	0.637	0.6250
R3-L	Weighted-F1	-0.35	0.457	0.4375
R3-L	ACC	-0.11	0.529	0.4375
R3-L	BWT	-2.92	0.0141	0.0625
R3-L	Forgetting	3.07	0.0105	0.0625
WACB	Accuracy	3.42	0.000899	0.0625
WACB	Macro-F1	7.56	0.000522	0.0625
WACB	Weighted-F1	2.46	0.00189	0.0625
WACB	ACC	3.06	8.58e-05	0.0625
WACB	BWT	-0.78	0.429	0.3750
WACB	Forgetting	0.90	0.347	0.3750
Freq	Accuracy	2.79	0.0017	0.0625
Freq	Macro-F1	6.13	7.31e-05	0.0625
Freq	Weighted-F1	2.12	0.00474	0.0625
Freq	ACC	2.53	0.000489	0.0625
Freq	BWT	2.03	0.0393	0.0625
Freq	Forgetting	-1.89	0.0439	0.0625
Replay-BS	Accuracy	3.47	0.0042	0.0625
Replay-BS	Macro-F1	5.49	0.001	0.0625
Replay-BS	Weighted-F1	2.39	0.00206	0.0625
Replay-BS	ACC	1.04	0.009	0.0625
Replay-BS	BWT	-0.85	0.373	0.3750
Replay-BS	Forgetting	0.98	0.302	0.3125
AR-Joint	Accuracy	-0.03	0.906	0.9375
AR-Joint	Macro-F1	-0.61	0.258	0.3125
AR-Joint	Weighted-F1	0.02	0.922	0.9375
AR-Joint	ACC	-0.08	0.444	0.5625
AR-Joint	BWT	0.71	0.129	0.1875
AR-Joint	Forgetting	-0.78	0.102	0.1250
ER-Bal-TabT	Accuracy	-3.36	9.71e-05	0.0625
ER-Bal-TabT	Macro-F1	-14.77	0.00264	0.0625
ER-Bal-TabT	Weighted-F1	-2.48	0.000128	0.0625
ER-Bal-TabT	ACC	-2.01	0.000143	0.0625
ER-Bal-TabT	BWT	4.01	0.241	0.4375
ER-Bal-TabT	Forgetting	-3.93	0.252	0.4375

61.41%, FPR 30.37%, and signed forgetting 16.93 pp. At the 1% calibration target, BAcc is 14.37 ± 0.12%, measured test FPR 1.04%, attack recall 0.35%, and forgetting 38.01 pp. At 2% and 5%, BAcc/FPR are 14.56/2.04% and 15.71/5.02%, respectively. High default false-alarm exposure and extremely weak low-FPR attack detection are both retained; completion of this block does not establish the CSE S1 family comparison. Across five seeds, checkpoint training totals 8855.73 s and summary elapsed time 9310.39 s; sampled peak child RSS is 9.910 GB (decimal). These elapsed measures overlap and are not additive. Each final memory has 1400 slots; 1171, 1179, 1182, 1180 and 1177 distinct class-feature pairs are not counts of unique originating row IDs. The original checkpoint hashes, source bindings and execution logs remain attached to the published group.

The completed CSE-2018 iCaRL adaptation has default BAcc48.95% at FPR60.44%, falling to BAcc28.46% at approximately0.97% TEST FPR under the 1% target. Its NME/softmax-KL adaptation is not the original sigmoid-BCE objective; feature membership does not prove unique originating row IDs. The completed ECBRS paper-memory block has default BAcc56.74% at FPR99.64%, and 1%-target BAcc14.35%; this is not a failure claim about the native binary method. CSE common-budget BiC has default BAcc 58.72 ± 13.18% at FPR58.39%; its 1%-target BAcc is 34.58 ± 8.04% at FPR0.9948%, attack recall23.93% and signed forgetting19.19pp. Weak low-FPR recognition and seed spread are retained. Each canonical completed group and its compatibility alias refer to the same five cells, not additional samples.

The archived CSE-2018 common-budget BiC group includes a disclosed checkpoint-resume limitation.[1]

[1]The derived resume record states `rng_restored=false`. Seed 14 resumed after six completed epochs with the saved model, optimizer and recorded epoch history, but the original checkpoint did not save or restore the global PyTorch CPU, NumPy, Python or DataLoader random-number-generator states. Deterministic per-epoch shuffling does not establish restoration of the full stochastic training trajectory. We therefore do not claim bit-exact equivalence to uninterrupted training. Seeds 10–12 retain their original Windows evaluations and seeds 13–14 their Linux evaluations; these historical results are not a homogeneous-runtime re-evaluation. The resumed attempt, reused completed work and interrupted partial work retain separate, potentially overlapping cost scopes.

Table 10: Completed added-baseline blocks and AR-Stateless reference, default decisions. Five seeds (10–14); BAcc is mean ± sample SD, other entries are means. Accuracy, BAcc, macro-F1 and FPR are percentages; signed forgetting is pp. Adapted implementations and access limits are described in the text.

Dataset	Method	Accuracy	Final BAcc	Macro-F1	FPR	Forgetting
UNSW	AR-Stateless	76.91	65.99 ± 0.66	51.17	1.26	27.42
UNSW	DER++	68.42	68.46 ± 0.61	43.67	24.05	28.23
UNSW	iCaRL (adapt.)	59.67	57.98 ± 1.31	35.10	33.71	28.19
UNSW	ECBRS (paper)	19.71	54.35 ± 0.45	20.95	98.63	41.26
UNSW	BiC (common flow)	65.93	66.71 ± 1.38	43.28	28.45	18.48
UNSW	ECBRS (author)†	60.74	66.20 ± 1.03	40.25	38.49	24.37
UNSW	BiC (native sched.)‡	67.32	68.50 ± 1.11	44.04	28.53	19.50
CIC-2017	AR-Stateless	99.44	95.72 ± 1.06	89.58	0.15	1.08
CIC-2017	DER++	90.97	92.35 ± 1.31	54.97	10.87	2.84
CIC-2017	iCaRL (adapt.)	56.43	73.77 ± 3.11	35.61	47.69	17.16
CIC-2017	ECBRS (paper)	25.55	70.79 ± 2.59	20.00	81.09	31.86
CIC-2017	BiC (common flow)	83.31	88.32 ± 6.24	48.69	18.72	7.92
CSE-2018	AR-Stateless	96.84	82.39 ± 0.96	69.25	1.38	15.79
CSE-2018	DER++	73.19	81.65 ± 2.06	61.41	30.37	16.93
CSE-2018	iCaRL (adapt.)	39.64	48.95 ± 3.32	22.83	60.44	20.78
CSE-2018	ECBRS (paper)	5.36	56.74 ± 0.37	16.12	99.64	49.88
CSE-2018	BiC (common flow)	44.48	58.72 ± 13.18	38.89	58.39	9.52

†UNSW-only author-memory sensitivity; not an additional primary cell or a native binary reproduction.
‡Historical native-stage-1 schedule sensitivity on flow inputs; excluded from the common-budget primary family and Holm comparisons. Not native packet reproduction.

Table 11: The same completed blocks at empirical calibration budgets. Final BAcc at 1% is mean ± sample SD; other entries are means. BAcc/FPR are percentages and signed forgetting is pp. Targets are not guarantees of test FPR.

		1% target			2% target		5% target	
Dataset	Method	BAcc	FPR	Forgetting	BAcc	FPR	BAcc	FPR
UNSW	AR-Stateless	65.41 ± 0.70	1.02	20.62	68.08	2.00	72.70	5.02
UNSW	DER++	12.17 ± 1.08	1.15	14.01	13.43	2.11	17.92	5.16
UNSW	iCaRL (adapt.)	12.90 ± 2.25	1.03	17.44	14.75	2.02	18.46	4.73
UNSW	ECBRS (paper)	11.78 ± 1.00	1.14	11.13	13.84	2.11	18.21	5.20
UNSW	BiC (common flow)	15.44 ± 2.38	1.03	13.97	17.59	2.01	23.10	4.83
UNSW	ECBRS (author)†	14.38 ± 2.50	1.07	7.38	16.57	2.14	21.37	5.13
UNSW	BiC (native sched.)‡	17.28 ± 2.75	0.84	12.04	19.89	1.93	25.63	4.82
CIC-2017	AR-Stateless	98.25 ± 1.09	1.00	0.42	98.61	1.99	99.10	4.93
CIC-2017	DER++	35.75 ± 11.24	0.99	24.15	46.19	1.95	79.37	4.87
CIC-2017	iCaRL (adapt.)	26.47 ± 3.32	0.94	8.49	29.40	1.92	36.63	4.90
CIC-2017	ECBRS (paper)	19.11 ± 4.13	0.98	14.52	21.19	1.96	23.72	4.90
CIC-2017	BiC (common flow)	35.87 ± 3.31	0.98	14.85	44.41	1.99	65.67	4.88
CSE-2018	AR-Stateless	81.72 ± 1.04	0.96	7.27	83.07	1.97	84.07	4.93
CSE-2018	DER++	14.37 ± 0.12	1.04	38.01	14.56	2.04	15.71	5.02
CSE-2018	iCaRL (adapt.)	28.46 ± 4.52	0.97	13.17	29.82	1.97	31.90	5.20
CSE-2018	ECBRS (paper)	14.35 ± 0.05	1.06	16.71	14.28	2.05	14.14	5.06
CSE-2018	BiC (common flow)	34.58 ± 8.04	0.99	19.19	34.92	1.99	40.50	4.98

Table 12: Added-baseline execution/publication coverage: 60/60 common-family primary cells. Each entry is completed seeds out of five; zero means no completed published seed, not a negative result. The five author-memory and five historical BiC native-schedule sensitivity cells are additional and excluded from this family.

Dataset	DER++	iCaRL	ECBRS	CER
UNSW	5/5	5/5	5/5	5/5
CIC-2017	5/5	5/5	5/5	5/5
CSE-2018	5/5	5/5	5/5	5/5

6.3 Registered S1 comparison on completed dataset blocks

Table 13 retains the original UNSW and CIC-2017 comparisons; Table 23 adds the completed CSE-2018 comparisons. Both test AR-Stateless (R3-A, exact C1), not R3-L, against DER++, the iCaRL adaptation, ECBRS paper-memory and common-budget BiC. Native-schedule BiC and author-memory sensitivities are excluded. The endpoint remains final BAcc at the original prior-shift 1% calibration target, with seeds 10–14 paired, one-sided Student t tests (df=4), and Holm correction over four baselines separately within each dataset. All twelve comparisons meet adjusted-$p < 0.05$. Differences range from 49.98 to 53.63 pp on UNSW, 62.38 to 79.14 pp on CIC-2017, and 47.14 to 67.37 pp on CSE-2018; the largest adjusted p is 1.53×10^{-4}. This closes that registered endpoint com-

parison, not the broader research gates or historical-runtime harmonization.

All eight unadjusted exact sign-flip values are 0.03125, the smallest attainable one-sided value from five pairs. Applying Holm to those sensitivity values would give 0.125, so multiplicity-corrected nonparametric confirmation is not established. The paired-t result relies on the seed-level difference assumptions, including approximate normality; five replicates offer limited diagnostics. These seeds share fixed, previously exposed datasets, not independent traffic populations. The comparison also retains the disclosed backbone and benign-refresh access differences, and is not a causal estimate of one architectural mechanism or a native-method replication.

Table 14 separates final Accuracy from stage-mean ACC and reports incremental BAcc and signed forgetting; default BAcc and FPR remain alongside these metrics in Table 10. S1 does not imply lower forgetting: on UNSW at 1%, AR-Stateless forgets 20.62 pp, compared with 14.01, 17.44, 11.13 and 13.97 pp for DER++, iCaRL, ECBRS and common-budget BiC, respectively. Its default forgetting is 27.42 pp versus BiC's 18.48 pp. The DER++ default-BAcc advantage and higher false-alarm burden stated above also remain. The two new tables come from the published `section178_s1` analysis, with per-seed metrics, raw confusion-count provenance and independent recomputation; no training was repeated. RQ1, RQ3 and the substantive-innovation gate remain open.

7 Deployment under strict false-alarm budgets

Low-FPR analysis is a separate deployment question, not the main default-decision ranking. Thresholds at 1%, 2% and 5% are selected using only the permanent Normal training holdout; actual test FPR is not guaranteed. The lead-reported developmental gate-versus-same-data-single-classifier final-BAcc gains at 1% are 17.0 points on UNSW and 24.8 on CIC (five seeds each, both reported $p < 0.01$). These results do not establish default-view superiority or isolate total computation. The original failed decisions are preserved; their detailed rules and chronology are in Appendix D.

7.1 Retention and plasticity under calibrated decisions

The historical checkpoint-calibrated analysis and CSE decision rounds are retained in Appendix B. Historical neural AR does not significantly improve final accuracy over Replay-BS on CSE18: lower signed forgetting accompanies approximately 4–5 points lower mean incremental balanced accuracy. These negative findings remain unchanged by the default-view presentation or the later design-space study.

7.2 Historical neural calibration-set operating points

Table 15 preserves the original operating-point snapshot at the common calibration target, including its then-available seed counts. It predates the supplemental Replay-BS seed 42; current five-seed CIC retention results are in Table 37. Historical operating-point tests and tables below are retained without recomputation.

On UNSW-NB15, AA-CIL obtains ADR 69.09%, Macro-F1 41.18%, and test FPR 1.92%. Relative to the strongest *external* ADR baseline, the gain is 18.67 percentage points. This comparison is descriptive with an associated paired test, distinct from the registered strongest-comparator conclusion.

AA-CIL's UNSW-NB15 Worms recall is 63.75%, with precision 3.52%; Fuzzers recall is 9.67%. The appendix reports the corresponding difficult-family metrics.

On NSL-KDD, AA-CIL reaches ADR 62.84%, compared with 69.67% for Freq. Its Macro-F1 is 55.66%, compared with 57.03% for WACB. Its test FPR, 6.15%, is substantially different from the calibration target. The method therefore neither wins across benchmarks nor delivers the nominal low-FPR operating point on this shifted test distribution.

At $\alpha = 2\%$, CIC-IDS-2017 AA-CIL has all five seeds: actual test benign FPR 1.93±0.02%, attack detection 83.66±2.83%, and Macro-F1 63.15±1.14% (mean ± sample SD). Against DER++ from the registered seven-comparator set, AA-CIL minus comparator ADR is -5.63±4.15 pp (one-sided $p = 0.98070$), which does not meet the dataset-level positive-difference and one-sided $p < 0.05$ condition. Against DER++ from the external-only comparator set, AA-CIL minus comparator ADR is -5.63±4.15 pp (one-sided $p = 0.98070$), which does not meet the dataset-level positive-difference and one-sided $p < 0.05$ condition.

7.3 Historical negative outcome

On UNSW-NB15, the strongest registered ADR comparator is AA-noAttMem, not Freq. AA-CIL's paired mean advantage is 1.15 percentage points, with one-sided $p = 0.0748$. It does not pass the registered significance criterion. NSL-KDD also fails the superiority requirement. Even a favorable CIC-IDS-2017 result cannot then provide superiority in the required majority of datasets. The primary registered criterion and conjunction are therefore negative; remaining individual criteria retain their actual complete or incomplete states. The criterion-by-criterion record is retained in Appendix D.1.

The separately reported external-baseline comparison does not alter this protocol's strongest-comparator definition or negative outcome.

7.4 What the ablations identify

On UNSW-NB15, TS-mem achieves ADR 24.79%, compared with 67.94% for AA-noAttMem. The latter uses

Table 13: Registered S1 at the 1% calibration target. Five paired seeds (10–14); difference is AR-Stateless minus baseline in percentage points. Paired one-sided t tests use df=4; Holm correction covers four baselines separately within each dataset. Exact sign-flip values are unadjusted sensitivity results, not a second decision rule.

Dataset	Baseline	Difference	t	Raw p	Holm p	Sign-flip p
UNSW	DER++	+53.25	93.13	3.98e-08	1.2e-07	0.03125
UNSW	iCaRL (adapt.)	+52.51	63.76	1.81e-07	3.62e-07	0.03125
UNSW	ECBRS (paper)	+53.63	280.68	4.83e-10	1.93e-09	0.03125
UNSW	BiC (common flow)	+49.98	40.55	1.1e-06	1.1e-06	0.03125
CIC-2017	DER++	+62.50	11.70	0.000153	0.000153	0.03125
CIC-2017	iCaRL (adapt.)	+71.78	41.03	1.05e-06	3.16e-06	0.03125
CIC-2017	ECBRS (paper)	+79.14	49.96	4.8e-07	1.92e-06	0.03125
CIC-2017	BiC (common flow)	+62.38	39.13	1.27e-06	3.16e-06	0.03125

Table 14: Descriptive companion metrics for the same seeds, including lower-forgetting baselines. All entries are seed means (percent or pp). Default final Accuracy and stage-mean ACC are distinct. At 1%, Inc. is stage-mean BAcc and Forget. is signed best-prior minus final old-class recall, including Normal. No tests are attached to these descriptive columns.

		Default				1% target		
Dataset	Method	Final Acc.	ACC	Macro-F1	Forget.	Final BAcc	Inc.	Forget.
UNSW	AR-Stateless	76.91	83.07	51.17	27.42	65.41	74.43	20.62
UNSW	DER++	68.42	76.98	43.67	28.23	12.17	35.67	14.01
UNSW	iCaRL (adapt.)	59.67	73.11	35.10	28.19	12.90	40.01	17.44
UNSW	ECBRS (paper)	19.71	57.78	20.95	41.26	11.78	34.08	11.13
UNSW	BiC (common flow)	65.93	75.82	43.28	18.48	15.44	42.38	13.97
CIC-2017	AR-Stateless	99.44	99.61	89.58	1.08	98.25	99.15	0.42
CIC-2017	DER++	90.97	94.95	54.97	2.84	35.75	67.82	24.15
CIC-2017	iCaRL (adapt.)	56.43	72.41	35.61	17.16	26.47	54.16	8.49
CIC-2017	ECBRS (paper)	25.55	55.84	20.00	31.86	19.11	47.38	14.52
CIC-2017	BiC (common flow)	83.31	91.66	48.69	7.92	35.87	64.99	14.85

greater benign refresh while retaining ordinary attack allocations.

The AA-CIL–AA-noAttMem contrast measures attack-memory transfer. Its UNSW-NB15 detection difference is small relative to the external-baseline gap and is not significant in the registered one-sided test.

7.5 Post-hoc exposure-matched and shift analyses

Replay-BS was added after the initial results as an exposure-matched control. Its single multiclass replay model uses the same refreshed benign samples, attack-only memory capacity, current attacks and calibration policy as AA-CIL.

On UNSW-NB15, AA-CIL improves attack detection over Replay-BS by 2.95 percentage points in the post-hoc paired test ($p = 0.0029$; Table 16). On NSL-KDD, the difference is 0.35 points ($p = 0.4052$), with higher test FPR and lower Macro-F1 than Replay-BS. The exposure-matched detection result therefore differs across datasets.

The diagnostic NSL sensitivity analysis contrasts training-domain benign-holdout thresholds with thresholds estimated using test-domain Normal labels. Test-Normal calibration brings the measured false-alarm rate to the target by construction but reduces attack detection and Macro-F1 for the evaluated gate variants (Table 18).

The post-hoc operating curves in Figs. 1–3 show the sampled false-alarm–detection and Macro-F1 trade-offs. Finite-grid points without a bracket remain unavailable; no extrapolation is applied.

7.6 Historical neural cost and storage

We charge the gate and family model separately and report their sum. Selection, calibration, evaluation and additional benign sample storage are separate quantities. Realized replay occupancy is reported alongside capacity because rare old classes may leave slots unused.

The permanent benign pool and the extra benign training inputs of AA-CIL, AA-noAttMem and Replay-BS are outside the replay ceiling but inside the disclosed total data footprint. TS-mem instead uses its retained Normal training rows within the ceiling. Normal validation rows already counted within TS-mem's replay allocation are not counted again as additional unique calibration rows. Successful optimization time excludes loading and serialization unless explicitly stated. Local GPU sharing and heterogeneous execution devices preclude interpreting raw runtime differences as an exclusive hardware benchmark. Failed, duplicate, or superseded executions are not scientific replicates and are accounted for separately from selected successful-cell costs.

Table 15: Final-task test performance at a benign-holdout calibration target of $\alpha = 2\%$. Rates are percentages; mean ± sample SD. Actual test FPR is not fixed at 2%.

Data	Method	n	Test FPR	Attack detection	Macro-F1	BAcc	Attack guard
NSL	Replay	5	4.08±2.16	65.39±5.30	46.83±4.93	47.05±4.52	34.83±5.78
NSL	DER++	5	2.51±0.45	65.28±2.73	50.46±3.15	50.11±2.97	38.27±3.77
NSL	WACB	5	3.62±0.87	66.83±3.77	57.03±3.91	56.69±5.22	46.77±6.49
NSL	Freq	5	4.93±2.42	69.67±2.53	56.84±4.32	61.82±2.44	53.50±3.50
NSL	Replay+LA	5	4.08±2.16	65.39±5.30	46.83±4.93	47.05±4.52	34.83±5.78
NSL	TS-mem	5	4.41±2.12	63.19±4.66	51.44±2.16	55.62±1.58	45.62±1.92
NSL	AA-noAttMem	5	4.32±1.82	60.29±3.16	56.67±3.34	56.97±2.69	47.29±3.05
NSL	AA-CIL	5	6.15±1.79	62.84±3.46	55.66±3.36	58.26±1.85	49.36±1.98
NSL	Replay-BS (post hoc)	5	3.65±2.16	62.48±4.64	57.05±3.33	59.32±2.85	50.07±3.92
NSL	Joint (offline)	5	6.79±1.00	66.63±1.80	59.37±2.90	58.65±2.20	50.01±2.75
UNSW	Replay	5	1.78±0.20	38.52±3.77	26.27±0.85	25.15±0.89	17.04±0.98
UNSW	DER++	5	1.98±0.27	18.48±4.12	14.47±1.44	14.43±1.03	5.15±1.16
UNSW	WACB	5	1.93±0.24	48.97±4.14	32.81±1.18	29.96±1.30	22.39±1.46
UNSW	Freq	5	2.01±0.26	50.42±2.93	31.22±1.73	28.93±1.40	21.25±1.54
UNSW	Replay+LA	5	1.78±0.20	38.52±3.77	26.27±0.85	25.15±0.89	17.04±0.98
UNSW	TS-mem	5	1.98±0.12	24.79±14.93	23.47±6.33	24.97±7.34	16.85±8.15
UNSW	AA-noAttMem	5	1.94±0.19	67.94±1.58	40.43±0.51	53.98±1.02	49.09±1.12
UNSW	AA-CIL	5	1.92±0.15	69.09±0.93	41.18±0.28	55.05±1.06	50.27±1.18
UNSW	Replay-BS (post hoc)	5	2.13±0.17	66.14±0.74	39.99±1.57	38.01±1.19	31.36±1.31
UNSW	Joint (offline)	5	1.93±0.15	67.11±1.45	37.22±1.00	35.38±0.67	28.42±0.76
CIC17	Replay	5	1.95±0.06	78.89±5.63	51.52±4.02	52.32±4.55	47.87±6.55
CIC17	DER++	5	1.92±0.07	89.29±3.01	48.85±3.19	48.44±2.90	47.68±3.63
CIC17	WACB	5	1.92±0.06	88.91±2.30	57.84±1.13	63.02±1.31	59.90±0.67
CIC17	Freq	5	1.93±0.02	88.55±2.32	57.97±4.33	64.03±2.62	61.81±5.97
CIC17	Replay+LA	5	1.95±0.06	78.89±5.63	51.52±4.02	52.32±4.55	47.87±6.55
CIC17	TS-mem	5	1.89±0.06	49.51±3.90	34.63±2.25	40.34±2.01	26.88±2.99
CIC17	AA-noAttMem	5	1.94±0.03	83.13±1.48	63.60±0.78	87.50±0.35	89.21±0.46
CIC17	AA-CIL	5	1.93±0.02	83.66±2.83	63.15±1.14	87.54±0.85	89.27±1.13
CIC17	Replay-BS (post hoc)	4	1.94±0.03	92.62±1.24	69.03±1.05	79.39±4.16	82.56±5.86
CIC17	Joint (offline)	5	1.93±0.04	98.19±1.94	68.43±1.30	78.68±3.54	78.56±3.58

Replay+LA and Replay share the same calibrated total logit adjustment and weights. Joint has offline old-training access. Replay-BS is post hoc. A dash means unavailable; $n < 5$ is descriptive only. The guard alone excludes attack classes with test support below 20.

CIC-IDS-2017 AA-family coverage: 15/15 cells across AA-CIL ($n = 5/5$), TS-mem ($n = 5/5$), and AA-noAttMem ($n = 5/5$); Replay-BS is reported separately. AA-CIL and AA-noAttMem refresh benign training rows outside replay memory; ordinary replay controls use quota-limited Normal training memory. Replay-BS matches the added benign exposure and attack allocation. Computational and causal limitations are discussed in Section 8.

8 Discussion and Limitations

8.1 Why AR-Stateless forgets more on UNSW-NB15

This is a post-hoc analysis of published confusion matrices, independently recomputed by `paper_aacil/check_forgetting.py`; it introduces no new training or candidate selection. Table 21 averages seeds 10–14 at the 1% calibration target. For each seed and old class, forgetting is the maximum recall before the final stage minus final recall; Normal is included and losses are not clipped. Taking the peak separately within each seed matters, particularly for Normal. The mean is 20.621 percentage points (pp), versus 9.506 for neural AR; the default-decision mean is 27.420 pp.

Exploits and DoS contribute 11.174 pp (54.19%) of the mean loss. At DoS introduction (stage 3), 62.77% of Exploits rows are predicted as DoS and Exploits recall falls to 18.38%, while DoS recall is 89.05%. At stage 4, when both are memory-only, recalls become 53.61% and 43.56%, respectively. This redistribution is consistent with confusion between the pair and the full-data advantage of the newly introduced class. It does not isolate feature overlap, sampling imbalance or finite memory as a causal explanation. Historical offline joint MLP results (Section 46, seeds 1–4/42) give final Exploits/DoS recalls of 86.37%/1.32%, but differ in learner, seeds and data access; they are not the matched joint-HistGB diagnostic or proof of irreducibility. Generic contributes another 1.827 pp; its historical joint final recall of 82.68% similarly supports a difficulty hypothesis, not an irreducible-error conclusion.

Fuzzers and Analysis contribute 5.531 pp (26.82%). Their final predictions are Normal for 86.07% and 67.33% of their test rows. These are real losses of old-attack recall under finite attack memory and refreshed benign data, not merely a reporting artifact. In the default view, Fuzzers falls from 88.99% to 11.33%, contributing 9.707 pp alone. Historical joint MLP Fuzzers recall is also low (4.09% at the 1% target). The observations support investigating finite-memory loss and benign–attack confusion, but cannot determine their separate causal shares or establish intrinsic "near-normal" families.

Lower forgetting need not imply better final retention: neural AR starts DoS, Reconnaissance and Backdoor at 27.83%, 67.28% and 34.00%, versus 89.05%, 98.55% and 76.57% here. Lower peaks leave less recall to lose. AR-Stateless has higher final recall for six of seven old attack classes; Exploits is slightly lower (50.83% versus 51.20%), not an established tie. These are descriptive comparisons, not per-class significance or equivalence tests. CIC-2017 contrasts with UNSW: mean forgetting is 0.417 pp and each old class's five-seed mean final recall is at least 98.25%. This does not prove absence of overlap or a bound for every individual seed.

Table 16: Ablations at $\alpha = 2\%$. Paired changes are AA-CIL minus the named ablation. Tests involving Replay-BS are post hoc; small groups do not amend registered decisions.

Data	Method	n	Detection (%)	MF1 (%)	Δ detection (pp)	$p_>$
NSL	AA-CIL	5	62.84±3.46	55.66±3.36	–	–
NSL	TS-mem	5	63.19±4.66	51.44±2.16	-0.35±4.01	0.5733
NSL	AA-noAttMem	5	60.29±3.16	56.67±3.34	2.54±3.42	0.0860
NSL	Replay-BS (post hoc)	5	62.48±4.64	57.05±3.33	0.35±3.09	0.4052
UNSW	AA-CIL	5	69.09±0.93	41.18±0.28	–	–
UNSW	TS-mem	5	24.79±14.93	23.47±6.33	44.30±15.74	0.0016
UNSW	AA-noAttMem	5	67.94±1.58	40.43±0.51	1.15±1.45	0.0748
UNSW	Replay-BS (post hoc)	5	66.14±0.74	39.99±1.57	2.95±1.23	0.0029
CIC17	AA-CIL	5	83.66±2.83	63.15±1.14	–	–
CIC17	TS-mem	5	49.51±3.90	34.63±2.25	34.14±5.78	0.0001
CIC17	AA-noAttMem	5	83.13±1.48	63.60±0.78	0.53±4.02	0.3908
CIC17	Replay-BS (post hoc)	4	92.62±1.24	69.03±1.05	-8.46±2.92	0.9949

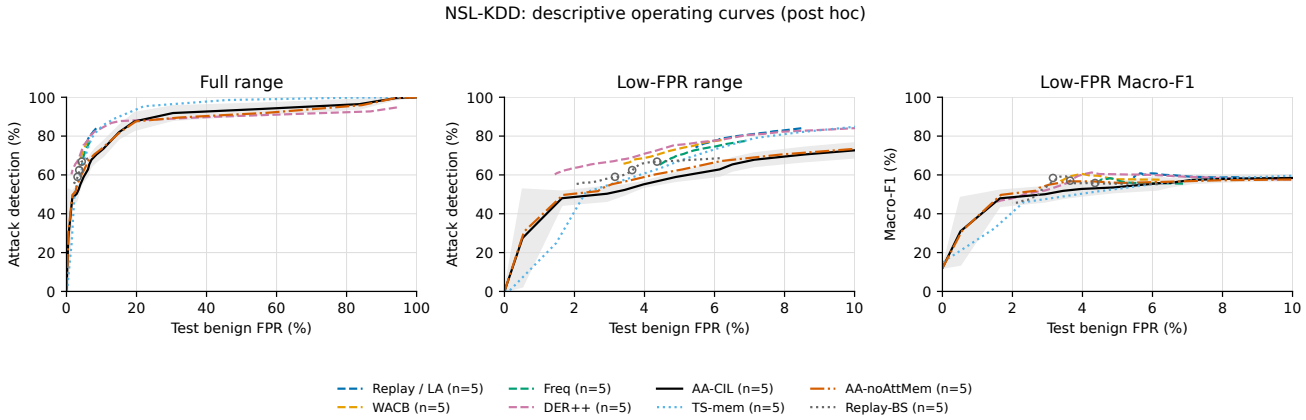


Figure 1: NSL-KDD operating curves (post hoc). Threshold-sweep summaries are descriptive, not calibration-set operating points or the registered test. Missing methods or seeds are identified in the generated panel.

Table 17: Post-hoc Replay-BS: single multiclass model with the AA-CIL benign exposure and attack replay allocation. This control was added after the original primary result.

Data	n	Test FPR	Detection	MF1
NSL	5	3.65±2.16	62.48±4.64	57.05±3.33
UNSW	5	2.13±0.17	66.14±0.74	39.99±1.57
CIC17	4	1.94±0.03	92.62±1.24	69.03±1.05

The post-hoc diagnostics are now complete for UNSW seeds 10–14 (20 runs): D1, the same HistGB trained on all seen-class optimization rows; D2, new attacks subsampled to the current memory share; and D3, attack-memory budgets multiplied by two and four. Permanent calibration rows remain excluded from optimization. Published Section 181 artifacts include checkpoint hashes, row-role checks, both decision views, and per-class first, peak and final recalls. Table 22 reports five-seed means; these diagnostics are not new candidates and do not change S1 or R1–R5.

D2 reduces measured forgetting from 20.62 to 7.53 pp while final BAcc changes from 65.41% to 64.51%. The reduction is about 64% of the original forgetting value, but is a descriptive between-condition contrast, not an identified causal fraction. It supports sensitivity of peak-minus-final forgetting to the full-data advantage at new-class introduction: reducing early recall peaks can lower forgetting without improving final performance. D2 changes both optimization and stopping samples, so the residual cannot be assigned entirely to finite memory. D3 supplies complementary memory-budget evidence: forgetting decreases monotonically at the evaluated budgets, while final BAcc increases. This supports memory sensitivity in this implementation, not a complete causal decomposition of forgetting.

D1 reaches final Exploits/DoS recalls of 83.30%/45.54% at the 1% target, compared with 50.83%/41.32% for AR-Stateless. Thus, the historical joint-MLP observation cannot justify an intrinsically inseparable-pair claim; we reject that interpretation. D1 is an offline reference, not a valid bounded-memory incremental competitor: it revisits old optimization data and also changes benign volume and old-class stopping access. AR-Stateless is 0.73 pp below D1 in calibrated final BAcc, but 7.42 pp below it in the default view, where D1's FPR is 14.26% rather than 1.26%. Neither the small calibrated gap nor the larger default gap should be reported without its decision policy and alarm burden. These are bounded RQ2 diagnostics; RQ1, RQ3 and the substantive-innovation gate remain unresolved.

8.2 Design-space study and negative variants

In the historical neural AR five-seed default-decision comparison, that candidate exceeds ER-Bal-TabT on UNSW by 1.6 points in accuracy and 2.7 in macro-F1,

Table 18: Post-hoc NSL-KDD calibration-distribution sensitivity at $\alpha = 2\%$. Test metrics are mean ± sample standard deviation in percent.

Method	Threshold calibration pool	n	Test FPR	ADR	MF1
AA-CIL	TRAIN Normal holdout	5	6.15±1.79	62.84±3.46	55.66±3.36
AA-CIL	TEST Normal (post hoc)	5	2.00±0.00	50.95±4.16	50.26±6.71
TS-mem	TRAIN Normal holdout	5	4.41±2.12	63.19±4.66	51.44±2.16
TS-mem	TEST Normal (post hoc)	5	2.00±0.00	53.55±3.31	47.81±1.00
AA-noAttMem	TRAIN Normal holdout	5	4.32±1.82	60.29±3.16	56.67±3.34
AA-noAttMem	TEST Normal (post hoc)	5	2.00±0.00	50.48±2.42	50.85±3.00

Each method uses the same paired seeds in both rows. TEST-Normal thresholds use test Normal labels and are evaluated on the same test population: this is a test-label-conditioned, non-deployable sensitivity analysis. The original superiority decision is unchanged; incomplete seed groups remain descriptive.

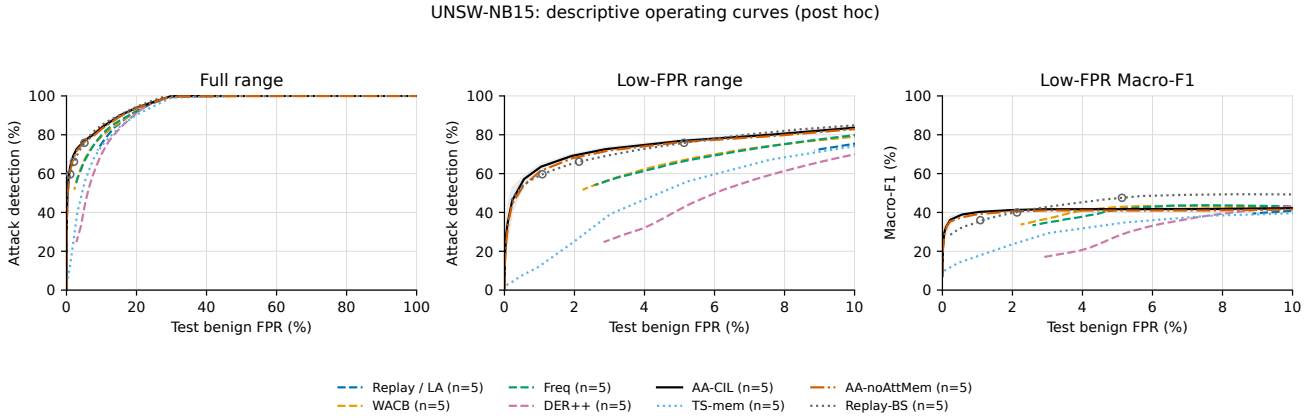


Figure 2: UNSW-NB15 operating curves (post hoc). Threshold-sweep summaries are descriptive, not calibration-set operating points or the registered test. Missing methods or seeds are identified in the generated panel.

but trails on CIC by 3.4 and 14.8 points. Under the low-FPR protocol neural AR trails ER on both datasets; these statements do not describe AR-Stateless. There is no overall winner across methods and views. These exploratory differences do not override the two-sided exact sign-flip sensitivity result ($p \geq 0.0625$ with five pairs).

The historical neural 48-configuration screen used seeds 1–2, with three selected configurations evaluated on seeds 10–14. None met K1 on either dataset; no configuration was confirmed in that screen, which was closed without another round of the same screen. Tables 44–45 preserve every screening configuration and the confirmation-stage comparators. T1/T2 were registered but not run (time), before scoring; they are not failed model results.

At the 1% calibration target, the leading selected configuration reaches final BAcc 51.2% on UNSW and 90.0% on CIC, versus ER-Bal-TabT's 56.3% and 92.7%. Its signed forgetting is 8.8 versus 25.0 points on UNSW, but 2.9 versus 2.7 on CIC: the forgetting advantage is not universal. AR-Joint is also retained as a negative variant (45.7% and 79.3% final BAcc). K2 passing means failure to detect significant worsening under its original rule, not demonstrated non-inferiority or equivalence. Among the selected UNSW configurations, forgetting ranges from about 8.1 to 11.6 points, so their similarity in final accuracy must not be generalized to every endpoint.

The historical neural candidate mechanism is asymmetric retention: refresh the separate benign–attack gate without retention regularization, while retaining old-attack knowledge in an attack-only family learner. DER-style replay, weight alignment and calibration are existing methods. The negative confirmation outcome neither establishes substantive novelty nor changes the locked research questions.

8.3 Historical neural comparisons: no universal winner

In the earlier gate/family experiments, the supported observation was dataset dependent: independently calibrated benign-refresh models substantially improved the selected external-baseline comparison on UNSW-NB15, whereas NSL-KDD did not improve the relevant registered endpoints. These observations concern the historical neural configurations, not the selected AR-Stateless learner. The evidence does not support a blanket superiority claim. In particular, favorable binary ADR can coexist with poor family discrimination: detected attacks assigned to the wrong family count as detected in ADR but not as correct in the multiclass metrics.

Those historical neural experiments also separate a design hypothesis from its identification. TS-mem shows the outcome of restricting benign refresh within a two-model architecture. The completed Replay-BS comparison strengthens the UNSW-NB15 evidence beyond an explanation based solely on additional benign exposure. Its NSL-KDD result does not support the same benefit, and its single network still receives less total model training than the gate-plus-family configuration. The supported claim is therefore an exposure-

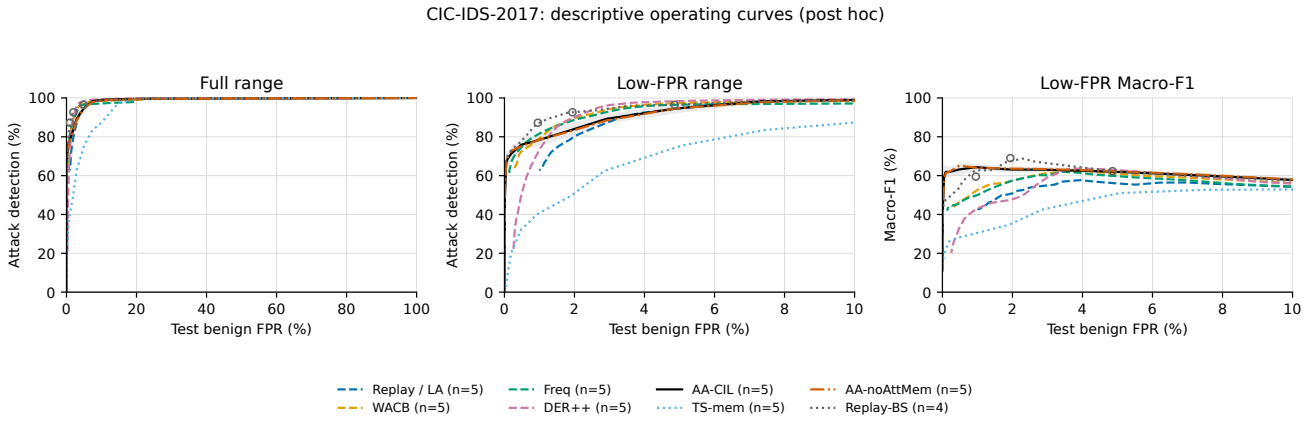


Figure 3: CIC-IDS-2017 operating curves (post hoc). Threshold-sweep summaries are descriptive, not calibration-set operating points or the registered test. Missing methods or seeds are identified in the generated panel.

Table 19: Descriptive final-task differences from the offline Joint reference at $\alpha = 2\%$. Values are AA-CIL minus Joint in percentage points; mean ± sample SD across paired completed seeds.

Dataset	Paired n	Δ FPR	Δ detection	Δ MF1
NSL	5	-0.641±1.101	-3.795±4.125	-3.706±5.262
UNSW	5	-0.006±0.118	1.981±1.719	3.959±0.990
CIC17	5	0.002±0.052	-14.531±2.836	-5.278±1.264

Joint uses offline old-training access and is neither a matched deployment baseline nor a guaranteed empirical upper bound. These gaps have no success threshold.

matched, dataset-dependent configuration benefit, not a compute-isolated causal estimate of decomposition.

The TS-mem–AA-noAttMem contrast changes benign exposure, realized gate updates and temporary storage together. The attack-memory contrast isolates a different allocation decision within that design, but its nonsignificant detection difference does not establish zero benefit or dispensability. Similarly, the gate/family decomposition partitions observed errors; it cannot by itself attribute forgetting to weight interference, because the operating threshold also changes between checkpoints.

In the retained original operating-point snapshot: The TS-mem comparison gives paired AA-CIL minus comparator differences of 34.14±5.78 pp in ADR and 28.52±1.74 pp in Macro-F1; the registered one-sided Macro-F1 test has $p = 1.67\times10^{-6}$. The AA-noAttMem comparison gives paired AA-CIL minus comparator differences of 0.53±4.02 pp in ADR and -0.45±1.76 pp in Macro-F1 (one-sided ADR $p = 0.39082$). The post-hoc Replay-BS comparison is incomplete (AA-CIL $n = 5/5$, Replay-BS (post hoc) $n = 4/5$); no complete-group inference is made. Regardless of this additional comparison, the failed original superiority requirement is unchanged. A CIC-only gain would not establish a universal advantage, and a post-hoc curve or Replay-BS contrast would remain distinct from that registered decision.

8.4 Calibration and deployment assumptions

Clean labeled benign data are a substantial resource. They may be contaminated, costly to curate, or distributionally unrepresentative of future traffic. Refresh from a fixed training pool is only a benchmark surrogate for continuous benign availability. It does not validate resilience to poisoned benign pools, temporal drift, network migration, or delayed labels. NSL-KDD's calibration-to-test shift illustrates why a nominal calibration target must not be equated with realized deployment FPR.

The test-Normal sensitivity analysis deliberately uses test labels and is not a deployable calibration rule or a replacement for the primary result. Threshold tightening exchanges false alarms for missed attacks, and same-population calibration does not establish out-of-sample reliability. Post-hoc mean curves at a common threshold and averages of per-seed interpolated ADR at a fixed FPR are different summaries; neither changes the holdout-calibrated protocol decisions.

8.5 Scope of statistical and architectural claims

The benchmark uses few paired seeds, fixed task orders, a finite known label universe, prepared tabular features, and a shared training implementation. These choices support controlled within-protocol comparisons but do not cover arbitrary task order, unseen network environments, or open-world label discovery. A matched schedule is not matched total computation or exposure. The historical fixed-capacity neural extension uses both UNSW and NSL for validation selection, so even its later five-seed results on these datasets are developmental, not blind confirmation. CSE-CIC-IDS2018 was withheld from this selection and assessed only after complete guarded release; it fails the registered conjunction. Heterogeneous local/HPC execution limits cost comparison, and withholding results does not es-

Table 20: Per-seed total training seconds across tasks and final memory occupancy. G and F are separately trained models; single-classifier methods have no G/F split. Timing excludes data loading/saving and is not an exclusive-GPU benchmark.

Data	Method	n	G (s)	F (s)	Total (s)	Used rows	G/BS benign MiB
NSL	Replay	5	–	–	31.0±0.4	852±0	–
NSL	DER++	5	–	–	70.0±1.0	986±0	–
NSL	WACB	5	–	–	30.4±0.2	927±0	–
NSL	Freq	5	–	–	31.3±1.0	784±0	–
NSL	AA-CIL	5	35.2±1.4	32.0±1.4	67.1±2.7	802±0	22.51±0.00
NSL	TS-mem	5	31.7±1.0	31.3±1.3	63.0±2.0	852±0	0.08±0.00
NSL	AA-noAttMem	5	34.7±0.8	30.6±1.1	65.3±1.8	652±0	22.51±0.00
NSL	Replay-BS (post hoc)	5	–	–	31.5±3.0	802±0	22.51±0.00
NSL	Joint (offline)	5	–	–	45.7±0.8	86±0	–
UNSW	Replay	5	–	–	34.7±0.4	1927±0	–
UNSW	DER++	5	–	–	80.0±1.6	1993±0	–
UNSW	WACB	5	–	–	34.2±1.1	2000±0	–
UNSW	Freq	5	–	–	33.6±0.7	1589±0	–
UNSW	AA-CIL	5	41.0±1.1	37.8±1.0	78.8±1.9	1905±0	13.35±0.00
UNSW	TS-mem	5	39.5±1.1	39.5±1.3	78.9±2.2	1927±0	0.05±0.00
UNSW	AA-noAttMem	5	40.5±0.6	37.9±0.4	78.4±0.9	1727±0	13.35±0.00
UNSW	Replay-BS (post hoc)	5	–	–	40.4±1.4	1905±0	13.35±0.00
UNSW	Joint (offline)	5	–	–	52.5±0.6	193±0	–
CIC17	Replay	5	–	–	219.8±6.0	1430±0	–
CIC17	DER++	5	–	–	553.9±13.2	1585±0	–
CIC17	WACB	5	–	–	223.0±3.8	1516±0	–
CIC17	Freq	5	–	–	221.2±3.5	1452±0	–
CIC17	AA-CIL	5	82.2±6.4	60.1±0.1	142.3±6.3	1402±0	15.07±0.00
CIC17	TS-mem	5	54.0±4.2	53.3±3.5	107.4±7.7	1430±0	0.05±0.00
CIC17	AA-noAttMem	5	70.9±2.8	53.0±1.9	123.9±4.3	1230±0	15.07±0.00
CIC17	Replay-BS (post hoc)	4	–	–	69.4±4.2	1402±0	15.07±0.00
CIC17	Joint (offline)	5	–	–	899.9±21.2	145±0	–

G/BS benign MiB reports final-task benign input feature-plus-index array bytes, not additional persistent storage. AA-CIL, AA-noAttMem and the single-model Replay-BS use up to 50,000 refreshed benign inputs outside M. TS-mem uses its retained training portion (normally 180 of 200 Normal slots) inside M; these bytes must not be added again to replay storage. Missing baseline byte records are shown as –, not zero. Dense-label materialization, permanent calibration features/indices, model weights and other transient copies are separate and excluded from this column; see the setup table. Joint is not a matched compute/storage baseline.

Table 21: UNSW AR-Stateless old-class recall and signed forgetting. Recall columns are percentages; loss is pp. Post-hoc, five-seed means.

Class	First	Pre-final peak	Final	Loss
Normal	99.032	99.108	98.979	0.129
Generic	99.354	99.354	84.739	14.615
Exploits	92.499	92.499	50.829	41.670
Fuzzers	36.405	36.405	9.493	26.912
DoS	89.048	89.048	41.324	47.724
Reconnaissance	98.551	98.551	88.824	9.727
Analysis	47.111	47.111	29.778	17.333
Backdoor	76.571	76.571	69.714	6.857

Table 22: Post-hoc UNSW retention diagnostics, seeds 10–14. BAcc and test FPR are percentages; signed forgetting (F) is in percentage points. The 1% column group is a calibration target, not a guarantee of test FPR. D1 is an offline reference with expanded old-data access.

	1% calibration target			Default decision		
Method	BAcc	F	FPR	BAcc	F	FPR
AR-Stateless	65.41	20.62	1.02	65.99	27.42	1.26
D2 matched exposure	64.51	7.53	1.02	64.60	4.64	1.05
D3 memory ×2	67.27	15.71	0.92	69.98	21.34	1.88
D3 memory ×4	67.46	12.03	0.96	71.77	17.25	2.77
D1 all-seen HistGB	66.14	2.85	1.05	73.41	3.93	14.26

tablish independent replication. Joint remains an offline reference whose available information differs from incremental methods.

The paired tests in the earlier neural studies discussed here are unadjusted; the later added-baseline comparisons use their separately registered within-dataset Holm correction. Selection of a comparator from the highest or lowest observed whole-group mean and the small seed count limit inferential strength. A nonsignificant harm test is not an equivalence or non-inferiority result. KD was not selected as the primary version by validation: its UNSW-NB15 reassessment is exploratory, and its NSL-KDD outcomes were already available before direct authorization. The first two NSL seeds also predate the reassessment document. We therefore do not describe NSL as blind confirmation or infer whether those available results had been inspected.

The initial manuscript window left the CIC-IDS-2017 KD group incomplete. A supplemental retention job reached the scheduler's `DEADLINE` state without allocation or execution; that job produced no scientific outcome. The subsequent deadline-free continuation completed all five KD and all five original AA-CIL trajectories. The later supplemental Replay-BS seed 42 closes its four-seed gap. Only the affected fixed-KD final-BAcc paired comparison was updated; other historical tests remain unchanged. Earlier operating-point tables and their associated interpretation retain their original four-seed Replay-BS snapshot, not a current missingness claim. Missing trajectories were missing evidence rather than negative model results.

The targeted literature review does not establish first-of-its-kind status. In particular, external benign data and attack-only old memory already appear in neighboring adaptation work. Our restricted-refresh ablation is not an implementation of that literature. Full-text unavailability for Wang and Cao further lim-

its precise differentiation. A novel substantive contribution would require verified distinction and an identifiable benefit beyond combining a gate, replay, and threshold calibration; neither architecture naming nor an external-baseline gain alone meets that standard.

8.6 Relation to the broader research questions

This manuscript addresses predictive retention, class-wise safety, and cost under class-incremental updates. It does not measure silent explanation drift across checkpoints, and it does not evaluate non-suppressive explanation governance. Those explanatory and governance questions remain unanswered by these experiments. Predictive gains alone cannot establish stable explanations or the safety of an explanation-governance mechanism.

8.7 Published score and operating-point diagnostics

Table 24 reports a post-hoc robustness check using one within-model attack-versus-Normal score and the permanent Normal holdout. Baseline BAcc changes in both directions, so no uniform improvement or model-mechanism explanation is inferred. At the common-score 1% target, the published paired-t/Holm comparisons against R3-A still reject all four baselines within each dataset; this is robustness evidence, not replacement of the registered original rule. Exact five-pair sign-flip tests have minimum one-sided $p = 1/32$ and cannot pass a four-test Holm family at .05. Newly evaluated baselines use the disclosed Linux runtime while the reference and original-rule evaluations remain as published; neither equal holdout targets nor this runtime arrangement implies equal measured TEST FPR or fully harmonized historical evaluation.

Table 25 uses TEST benign labels after evaluation and is an unregistered NSL-KDD extension, not a deployable threshold-selection rule or S1 decision. At identical 194/9711 false positives per seed (1.9977%), neural AR has mean BAcc 51.27%, above AR-Stateless at 47.44%; R3-L is numerically close at 47.45%. AR-Stateless therefore does not show an all-method matched-FPR clean win. Under default decisions it has the lowest BAcc among these ten methods. At the original 1% holdout target, measured TEST FPR is 6.1% for AR-Stateless and 1.4% for DER++; this is calibration transfer, not matched actual FPR. At the 5% budget its mean actual FPR is 4.6298%, versus 4.9943% for the other methods, so that comparison is not strictly matched. The original default and holdout-target results remain unchanged.

Table 26 shows benign holdout/TEST KS of 0.006784–0.010403 and measured TEST FPR near 1%, alongside WACB BAcc of 21.40–72.65% (mean 42.19%, sample SD 22.33 pp). These associations do not identify a feature-level cause, a memory mechanism, or a bisection failure. Stored calibration searches were monotone and budget-bracketed; independent raw-logit calculations reproduce all twenty original TEST CMs and fifteen holdout FP counts. The diagnostic uses forward batches4096 versus original configuration 2048; CM/FP agreement does not prove original logits bit-exact. Scores have model-specific scales and must not be compared as absolute common units across seeds. The successful read-only job used 87 s wall/87.914 CPU seconds; including the earlier owner-cancelled attempt and startup-only diagnosis gives 1153 s sequential HPC wall/99.854 CPU seconds, not a training speedup. The earlier stall did not recur but its cause remains unproved. Exact-byte split arrays, restoration code and completed logs accompany the published evidence; import-time stderr is nonempty.

Table 23: Completed CSE-2018 registered S1 companion to Table 13. Original prior-shift rule, five paired seeds, one-sided t and Holm over four baselines. Differences are pp; exact sign-flip sensitivity remains unadjusted.

Baseline	Difference	t	Holm p	Sign-flip p
DER++	67.34	152.86	2.2e-08	0.03125
iCaRL (adapt.)	53.26	24.90	1.54e-05	0.03125
ECBRS (paper)	67.37	145.44	2.2e-08	0.03125
BiC (common flow)	47.14	13.24	9.4e-05	0.03125

9 Conclusion

The selected AR-Stateless candidate improves final and mean incremental balanced accuracy over ER-BalTabT on UNSW and CIC-2017 at the registered 1% calibration target. Its R3 harm tests do not detect higher forgetting against that comparator; this is not equivalence or non-inferiority. The CSE margin over R3-L passes the registered t test narrowly but not the sign-flip sensitivity check at .05, and forgetting exceeds neural AR and R3-L. Added-baseline coverage is now 60/60, with all twelve original-rule S1 paired-t/Holm comparisons passing across three datasets; exact sign-flip sensitivity does not establish multiplicity-corrected nonparametric confirmation. DER++ retains its UNSW default-BAcc advantage at higher FPR. Weak low-FPR adaptations, heterogeneous historical evaluation and the BiC RNG-resume limitation remain explicit. The NSL TEST-informed matched 2% diagnostic favors neural AR over AR-Stateless, so the registered results do not imply universal superiority. These conclusions address the disclosed RQ2 setting, not RQ1, RQ3 or completion of the substantive-innovation requirement. Class-incremental NIDS must balance false alarms, attack detection and retention as new families arrive. The earlier neural comparisons support a trade-off study: retention components can reduce forgetting while impairing learning of new classes. Their historical conclusions follow and do not describe the later AR-Stateless candidate. Benign-stream training with an independently calibrated gate supports strong attack detection in the studied setting, but the original decoupled model exhibits 20.04 and 11.12 pp forgetting on UNSW-NB15 and NSL-KDD. The validation-selected CBF variant failed the retention criterion; KD reduced

Table 24: Post-hoc common-score robustness at the 1% holdout target. Each entry is original prior-shift → common-score final BAcc (five-seed mean, percent). This does not replace the registered primary or match measured TEST FPR. New baseline forward evaluations use Linux; historical reference/prior results remain as published.

Dataset	DER++	iCaRL	ECBRS	BiC common
UNSW	12.17→15.88	12.90→18.69	11.78→17.55	15.44→21.52
CIC-2017	35.75→33.20	26.47→26.33	19.11→20.99	35.87→40.57
CSE-2018	14.37→15.41	28.46→27.47	14.35→20.22	34.58→32.02

Table 25: Unregistered NSL-KDD TEST-label-informed post-hoc operating points, five-seed means (percent). At 2% all methods have 194/9711 false positives per seed. At 5%, AR-Stateless has a lower actual FPR, so that column is not strictly matched. No new significance decision is attached.

Method	2% BAcc	5% BAcc	5% actual FPR
AR-Stateless	47.44	68.79	4.63
R3-L	47.45	67.17	4.99
Neural AR	51.27	56.15	4.99
Replay-BS	43.25	59.01	4.99
freq	43.29	57.40	4.99
wacb	43.80	64.67	4.99
derpp	42.63	62.96	4.99
icarl	34.52	39.57	4.99
ecbrs	37.70	53.42	4.99
bic	39.30	43.66	4.99

Table 26: CSE-2018 WACB saved-model score diagnosis. Original 1% prior-shift rule; BAcc/FPR are percent. KS compares benign holdout and TEST scores within each model, not across models.

Seed	BAcc	TEST FPR	Benign KS
10	21.40	0.9972	0.007459
11	24.73	1.0467	0.009446
12	33.86	0.9876	0.006784
13	58.33	0.9844	0.009052
14	72.65	1.0033	0.010403

forgetting to 2.31 and 0.98 pp with attack-detection rates of 70.53 and 71.67% at nominal $\alpha = 2\%$. Nevertheless, its final balanced-accuracy advantage over the matched-exposure Replay-BS control on NSL-KDD was not significant ($p = 0.1329$), while the NSL test FPR of 7.94% exposes calibration shift. These results support a trade-off study, not a fully resolved false-alarm–forgetting–accuracy solution. KD was not the validation-selected main version; UNSW is exploratory and NSL is not blind confirmation.

On the previously withheld CSE-CIC-IDS2018 assessment, the fixed KD+WA candidate reaches 72.69% final balanced accuracy, 5.27 pp signed forgetting and 95.44% attack detection at 1.95% test FPR. Its final accuracy is 7.06 pp below the exposure-matched single classifier and 5.85 pp below original AA-CIL. The accuracy-superiority and two-point accuracy-cost conditions fail, while the forgetting and attack-detection harm tests do not detect significant harm. This is not equivalence or proof that retention is solved. No test result changes the selected configuration.

The historical neural AR round does not reverse that negative history. In the lead-reported seeds 5–9 summary, AR's final BAcc is descriptively close to Replay-BS at 2%, without demonstrated superiority or equivalence. At 1%, the 1.9pp advantage is not significant: H-low's external-comparator half passes for AR, while its Replay-BS half fails. The roughly 4–5pp mean incremental BAcc cost and lower signed forgetting must be reported together. The lowest-forgetting statement is limited to the five available decision methods, not all variants; the complete nine-method descriptive extension is reported separately in Table 41 without changing that historical decision. The later AR-Stateless study uses its separately registered criteria and seeds 10–14; it does not revise the earlier candidate's outcome.

The resulting contribution is a reproducible study of this unresolved trade-off under an explicit benign-data availability assumption. Clean benign access, calibration shift, unmatched total computation and the CIC-IDS-2017 accuracy cost of retention limit generalization. Silent explanation drift and non-suppressive explanation governance remain separate, unanswered research questions.

Reproducibility and evidence availability

Earlier tables and numerical prose are generated from repository result files. The explicitly labelled historical neural AR new-seed follow-up tables and their associated statements transcribe the lead's rounded recomputation in work-order Sections 71–72, except the 2% ADR and test-FPR columns averaged from the five saved final values per method in the published seed releases. The developmental strict-FPR gate gains in the deployment analysis are attributed to the lead's report in Section 73.1; this editorial update does not rerun their tests. In contrast, the later AR-Stateless confirmation and added-baseline tables are generated from their published result files and confusion counts, with their own evidence revisions and seed blocks. The companion code retains training configurations, seeds, taskwise confusion matrices, sample-access policies, and separate gate/family costs. Model payloads and datasets are not embedded in the manuscript. Evidence cutoffs and actual group sizes apply to their respective rounds; unfinished results are not inferred from a running process or an anticipated completion time.

A Historical neural design-space study

This appendix preserves the earlier AA-CIL gate/family design and its retention extensions. None of these neural architectures, losses or update algorithms describes AR-Stateless. Their original data roles, development selection rules and negative outcomes are retained; relocation does not reclassify exposed evidence as independent confirmation.

A.1 Independent detector and family learner

The gate is a binary MLP trained on

$$D_t^G = B_t \cup D_t^A \cup R_{t-1}^{A,\mathrm{tr}}. \tag{4}$$

Its labels are benign versus attack. The family MLP is trained on

$$D_t^F = D_t^A \cup R_{t-1}^{A,\mathrm{tr}}, \tag{5}$$

using attack-family labels only. Both use the existing class-balanced cross-entropy implementation and retain learned parameters between updates. Their optimizers and costs are separate. Neither model receives calibration data as optimization input.

For exact implementation fidelity, the family network is initialized with the benchmark's full attack-label output dimension. Training follows the shared replay cross-entropy implementation, while inference restricts the argmax to observed attack families. Thus this experiment assumes a known finite benchmark label universe; it is not a claim about an unbounded, dynamically expanding unknown-label vocabulary.

Algorithm 1 One AA-CIL update

Require: Current attacks D_t^A; retained attack memory R_{t-1}^A; benign pools $B^{\mathrm{tr}}, B^{\mathrm{cal}}$

Sample B_t from B^{tr} only.
Train G on $B_t \cup D_t^A \cup R_{t-1}^{A,\mathrm{tr}}$.
Train F on $D_t^A \cup R_{t-1}^{A,\mathrm{tr}}$.
Allocate attack capacities $q_{t,c}$ within M_t.
for each observed attack c **do**
    Select new-class rows, or subsample retained old rows; keep training and validation disjoint.
end for
Calibrate $q_{t,\alpha}$ on B^{cal}; do not update weights.
Save models, retained memory, allocation, costs, and operating-point confusion matrices.

A.2 Attack replay allocation

Let $K_t = |\mathcal{A}_t| + 1$, including Normal, and let m be the common per-observed-class budget. The shared capacity is

$$M_t = mK_t, \qquad m = 200. \tag{6}$$

AA-CIL divides M_t among observed attacks: $q_{t,c} = \lfloor M_t/|\mathcal{A}_t| \rfloor$ plus one for the first remainder classes in ascending class-ID order. This is a capacity ceiling, not a promise of equal actual occupancy. Each class allocation contains separate training and validation slots; validation rows do not enter optimization. When a class is new, examples can be selected from its prepared training and validation partitions. Once it is old, the corresponding retained arrays are the only permissible source, even if a later quota exceeds available rows. Unused slots remain unused; old pools are never revisited to fill them.

This allocation transfers the nominal Normal share to attacks. It does not make benign data free in a physical-storage sense: both benign training samples and the permanent calibration pool are disclosed outside M_t. The no-transfer ablation tests whether this allocation adds value beyond benign refresh.

A.3 Calibration and inference

Let $s_G(x) = z_A(x) - z_B(x)$ be the gate's attack-minus-benign logit margin. For $n = |B^{\mathrm{cal}}|$, sort its margins as $s_{(1)} \le \cdots \le s_{(n)}$ and set

$$k_\alpha = \lfloor \alpha n \rfloor, \qquad q_{t,\alpha} = s_{(n-k_\alpha)}. \tag{7}$$

The strict decision $s_G(x) > q_{t,\alpha}$ bounds the empirical calibration false-alarm count by k_α, including ties. No randomized tie breaking is used. Prediction is

$$\hat{y}(x) = \begin{cases} \text{Normal}, & s_G(x) \le q_{t,\alpha}, \\ \arg\max_{c \in \mathcal{A}_t} F_c(x), & s_G(x) > q_{t,\alpha}. \end{cases} \tag{8}$$

This is an empirical training-holdout operating point, not a distribution-free guarantee on test or deployment FPR. Gate false negatives cannot be recovered by the downstream family classifier.

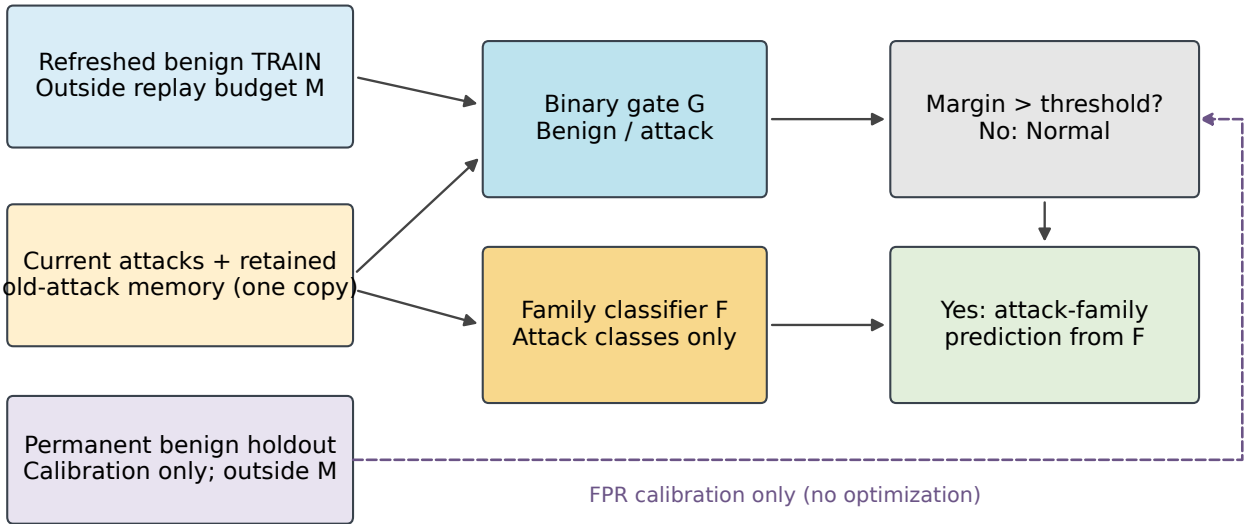


Figure 4: AA-CIL separates detection from family classification. The calibration pool never enters training. A single retained attack memory supplies both learners and is counted once; two networks and both training schedules are counted in cost.

A.4 Post-hoc retention extension: AA-CIL-KD

The original AA-CIL gate relies on attack replay without an explicit response-retention loss, and its family learner uses ordinary replay. AA-CIL-KD is a post-hoc extension to that study, with its protocol and success criteria registered before observing any KD results. It adds a gate loss

$$\mathcal{L}_G = \mathcal{L}_{\mathrm{CE},G} + \lambda_G \, \mathcal{L}_{\mathrm{MSE},G}. \tag{9}$$

For retained old attack training rows, the gate targets are the stored two-logit responses recorded when those rows entered memory. For the current benign sample, targets come from the frozen preceding-task gate. New attack rows have no old-class distillation target, and the first task has no preceding-task gate. Neither calibration rows nor attack validation rows become optimizer inputs.

The attack-only family learner uses a task-boundary DER++-objective adaptation: supervised learning is supplemented by replay cross-entropy and MSE against stored family logits. This is not a claim to reproduce the original DER++ online storage schedule. Gate and family logits are auxiliary records for the same attack rows, not two separately allocated sample memories; their storage and the two training costs must be disclosed. The update budgets, benign refresh, permanent holdout and threshold policy remain those of AA-CIL.

An initial development search used mean stagewise validation balanced accuracy and varied three distillation coefficients. It was superseded before formal KD results by the retention-constrained selection below. Its completed runs remain part of the development cost, but do not enter the new selection. The previously planned noGKD ablation is not one of the five family-model candidates in this revised experiment.

A.5 Post-hoc parameter isolation: AA-CIL-ISO

AA-CIL-ISO changes only the family learner relative to AA-CIL-KD: the gate, G-KD objective, benign refresh and calibration remain the same. It uses the existing MLP feature encoder. On the first task containing attacks, the encoder and that task's new family heads train jointly within one family-model update budget. The encoder is then frozen. Each new family c receives its own rank-8 residual adapter and two-logit binary head,

$$h_c = h + U_c V_c h, \qquad p_c(x) = \mathrm{softmax}(H_c h_c)_1. \tag{10}$$

The head learns that family versus the other attacks available in the current task and legal old-attack replay. Multiple new families provide negatives for one another. Normal rows are not used as family-head negatives. After the task, its adapters and heads are frozen permanently; old attack pools are never reopened.

All heads introduced in one task share a single training loop: their class-balanced binary cross-entropies are averaged before each optimizer step. Each new head participates in the task's common $\max(8 \text{ epochs}, 2000 \text{ steps})$ schedule, rather than receiving an additional serial budget per head. The first encoder training has no separate pretraining budget. Joint family-loop time and shared steps are recorded once; exclusive per-head time is not inferred from that total. If the first task contains only one attack family, there are no other-attack negatives: this limitation is recorded explicitly, without invented negatives or benign substitution.

After the gate declares an attack, prediction is $\arg\max_{c \in \mathcal{A}_t} p_c(x)$ over independently normalized positive probabilities. This is head-only inference with no router and no cross-head softmax calibration. It adapts

OFRA-style parameter isolation to the present backbone and is evaluated under the validation-selected retention protocol, with descriptive paired comparisons against KD.

A.6 Retention alternatives and validation-only selection

The revised experiment compares five family learners under the same G-KD gate: KD, ISO, weight alignment (WA), class-balanced fine-tuning (CBF), and nearest-class-mean prediction (NCM). WA rescales newly introduced classifier weights by the mean old-to-new weight-norm ratio after task training [13]. CBF adds small-learning-rate updates on balanced retained attack training memory [15]. NCM computes class prototypes from retained attack training rows encoded by the current family encoder and predicts by feature-space distance [17]. These are adaptations of established mechanisms, not claims of novel versions of the cited algorithms. Their additional time, steps, prototype storage and model storage are reported separately.

Only UNSW-NB15 development seeds $\{5, 6\}$ select hyperparameters and the main version. Each candidate uses $\lambda_G \in \{0.5, 1, 2\}$; KD fixes both family DER coefficients to one. CBF uses 0.1 times the base learning rate and either one or five fine-tuning epochs. A stratified permanent 10% attack holdout supplies fixed validation cohorts across stages; it is never optimized, is accessible only for this development evaluation, and its storage is disclosed outside the replay budget. The permanent Normal holdout supplies the gate threshold at $\alpha = 2\%$ and the Normal validation cohort. This development-only access exception does not reopen old attack training pools or change formal replay access.

For each version, eligible configurations have final validation balanced accuracy at least the original AA-CIL development mean minus three percentage points. The configuration with lowest mean validation forgetting is retained, and the lowest-forgetting eligible version is selected as the candidate main method. All choices are frozen across datasets before evaluation seeds $\{1, 2, 3, 4, 42\}$ are used. Every version is reported, not only the selected one. If no configuration is eligible, there is no eligible main method; any configuration retained for descriptive formal reporting is explicitly a fallback, not an eligibility success. Test labels never select the gate threshold, hyperparameters, or main version.

A.7 Fixed-capacity retention candidates

A subsequent extension constrains the family learner to a fixed parameter count across tasks, no more than twice the full original family model, at most one extra component, and a single inference model. KD-weak reduces the family DER coefficients; KD+WA and KD+CBF apply alignment or balanced fine-tuning after KD. EMA-KD maintains one slow family copy (decay 0.999), uses replay-logit MSE with weight one, and predicts only with that copy. DUAL uses two fixed encoders and a joint concatenation head: the stable encoder has the frozen KD coefficient, while the plastic encoder has no distillation loss. The all-attack output capacity is allocated initially; unseen classes remain masked until introduction. There is no taskwise branch expansion.

The fixed-capacity retention protocol uses both UNSW-NB15 and NSL-KDD development seeds $\{5, 6\}$ with the fixed attack validation cohorts and permanent Normal holdout. It requires each dataset's validation forgetting to be at most its lowest external-five value plus one percentage point before comparing mean validation final BAcc. Within one point of BAcc, the simpler candidate is preferred. Gate coefficients are $\lambda_G \in \{0.25, 0.5\}$; KD-weak searches each DER coefficient in $\{0.25, 0.5\}$, and the other KD configurations retain coefficients of one. CBF uses one epoch at 0.1 times the base learning rate. If no candidate meets both forgetting constraints, the lowest-mean-forgetting version is frozen only as an explicitly ineligible descriptive fallback. All five candidates are reported on formal seeds, with UNSW and NSL still designated development datasets.

B Historical checkpoint-calibrated retention

B.1 Class-incremental retention and forgetting

Class-incremental performance is evaluated before the false-alarm–detection comparison. For every saved checkpoint $t = 1, \ldots, T$, the operating threshold is determined anew from the same permanent Normal calibration pool at $\alpha = 2\%$, using that checkpoint's scores. The final threshold is not retrospectively applied to earlier models. For a seen class c, let $r_{c,t}$ denote its recall under this operating point and let $\mathcal{C}_t$ be the seen-class set. We report

$$B_t = \frac{1}{|\mathcal{C}_t|} \sum_{c \in \mathcal{C}_t} r_{c,t}, \qquad \overline{B}_{\text{inc}} = \frac{1}{T} \sum_{t=1}^{T} B_t, \tag{11}$$

together with final balanced accuracy B_T. Mean incremental balanced accuracy weights stages equally, rather than pooling their examples or reporting micro accuracy.

Let τ_c be the introduction stage of class c and $\mathcal{O} = \{c : \tau_c < T\}$. Signed final forgetting is

$$\mathcal{F} = \frac{1}{|\mathcal{O}|} \sum_{c \in \mathcal{O}} \left(\max_{\tau_c \le t < T} r_{c,t} - r_{c,T} \right). \tag{12}$$

The historical maximum excludes the final stage; negative values therefore retain final improvement and are not clipped to zero. The primary definition includes Normal whenever it belongs to $\mathcal{O}$; an attack-only version excludes Normal from this same set. Classes introduced in the final task have no subsequent update and are not assigned a forgetting value. No rare-class support filter is applied to these Class-IL summaries:

the support filter belongs only to the separate attack-family safety guard. We report per-seed values and their mean and sample standard deviation, with forgetting in percentage points. Low forgetting alone is not a ranking of useful retention: a class that was never learned well has little recall to lose. It must be read together with final and mean incremental balanced accuracy for every method. Incomplete trajectories or seed groups remain explicitly incomplete.

The completed original-AA trajectories show why detection and retention must be separated. On UNSW-NB15, mean incremental BAcc is 69.65%, final BAcc is 55.05%, and signed forgetting is 20.04±1.16 percentage points. On NSL-KDD, the corresponding values are 74.24%, 58.26%, and 11.12±2.70 points. A higher trajectory average therefore coexists with substantial old-class recall loss; the new KD and ISO variants target that unresolved endpoint rather than converting the earlier detection gain into a retention claim.

For each old attack class, the confusion matrix partitions its error into a gate miss (prediction Normal) and a wrong attack-family prediction. At the same class-specific pre-final peak used above, the recall loss equals the increase in these two error rates. Both components use the same class support and peak checkpoint.

The validation-selected retention protocol makes retention mandatory on both UNSW-NB15 and NSL-KDD. Its selected version must satisfy the forgetting, final-BAcc and ADR conditions in Appendix D.2. CIC-IDS-2017 is supplementary to this protocol. Mean incremental BAcc, Macro-F1 and actual test FPR accompany all five variants.

UNSW training-validation seeds 5/6 locked AA-CIL-CBF ($\lambda_G = 2$, 1 CBF epoch(s), learning-rate multiplier 0.1) as the eligible main version. AA-CIL-KD, AA-CIL-ISO, AA-CIL-NCM are reported as development-ineligible fallback configurations. Complete formal five-seed groups: UNSW 5/5 versions, NSL 5/5 versions. On UNSW-NB15, AA-CIL-ISO recorded signed forgetting of 0.018±0.032 pp alongside final balanced accuracy of 19.67±0.02%. ISO minus KD differences were -2.29±1.03 pp for forgetting ($p_< = 0.00385$) and -18.86±1.10 pp for mean incremental balanced accuracy ($p_< = 1.37 \times 10^{-6}$), using five paired seeds and unadjusted one-sided tests. The validation-selected retention protocol fails a necessary condition; the false-alarm–forgetting–accuracy trade-off remains unresolved.

Separately scoped KD reassessment. The fixed-KD reassessment uses $\lambda_G = .5$ and DER $\alpha = \beta = 1$. On NSL-KDD, the final BAcc difference from Replay-BS is 1.49±2.58 pp ($p_> = 0.1329$); the required superiority condition is not met. A necessary numerical condition fails; the trade-off remains unresolved. Across all five CIC17 KD seeds, mean incremental BAcc is 87.59±1.07%, final BAcc is 79.20±4.08%, and signed forgetting is -10.54±9.15 pp. At the nominal 2% calibration target, test FPR, ADR, and Macro-F1 are 1.93±0.04%, 92.10±2.84%, and 65.94±2.49%, respectively.The CIC-IDS-2017 KD group is complete (5/5 seeds).

Fixed-capacity retention extension. The fixed-capacity retention protocol completed validation-only selection at 2026-10-20 03:17:14.796356+08:00. None of the five candidates satisfies the forgetting constraint on both development datasets. KD+WA is retained only as an ineligible descriptive fallback with the lowest average signed validation forgetting, with $\lambda_G = 0.5$ and family DER coefficients $\alpha = 1$, $\beta = 1$. This selection did not meet the retention requirement and does not resolve the trade-off. 10 of ten UNSW/NSL five-seed candidate/dataset groups are complete. The additional CIC17 KD+WA five-seed group is descriptive, not part of validation selection. UNSW, NSL and CIC17 remain development datasets. The separately released CSE-CIC-IDS2018 assessment is reported below; it does not change the frozen selection.

Figure 5 separates predictions of Normal from wrong attack-family predictions at the final task. It does not measure forgetting without the preceding checkpoints and does not change the failed CSE assessment.

CIC-IDS-2017 development accuracy cost. On CIC-IDS-2017 the retention components remove forgetting but cost 8.3 pp final BAcc and give no gain over the same-data single classifier. Here "remove forgetting" refers to the negative group-average signed forgetting endpoint, not zero loss for every class or proof of a causal retention mechanism. The 8.3 pp cost is relative to original AA-CIL; the same-data single classifier is Replay-BS, which matches exposure but not total computation. This descriptive comparison neither reselects the fixed candidate nor changes the failed historical protocols.

B.2 Withheld CSE-CIC-IDS2018 assessment

On the previously withheld CSE-CIC-IDS2018 assessment, the fixed KD+WA candidate reaches 72.69% final balanced accuracy, 5.27 pp signed forgetting and 95.44% attack detection at 1.95% test FPR. Its final accuracy is 7.06 pp below the exposure-matched single classifier and 5.85 pp below original AA-CIL. All 35 physical reference cells and five selected-candidate cells completed before the guarded releases. The four saved decisions are pass, fail, fail, pass. Criterion 1 uses Freq, the lowest-mean forgetting member of the fixed external five. Paired one-sided t-tests determine the registered significance conditions; exact sign-flip values enumerate all 32 assignments as sensitivity checks. Criterion 3 uses only the mean difference, not a significance test. Reference seeds 2/42 and selected seed 42 ran locally; the other cells ran on HPC A100. Costs retain this hardware distinction. The proposed new-attack mask is already present in the executed KD+WA code: current new attacks receive binary CE only, and gate KD applies to benign and replayed old attacks. The proposed duplicate variant was therefore not trained; any additional intervention requires a genuinely dis-

Table 27: NSL: checkpoint-calibrated Class-IL performance at $\alpha = 2\%$.

Method	n	Mean inc. BAcc	Final BAcc	Forget (all old)	Forget (old att.)	Final ADR	Final MF1
Replay	5	70.12±2.82	47.05±4.52	9.73±4.30	12.53±5.85	65.39±5.30	46.83±4.93
Replay+LA	5	70.12±2.82	47.05±4.52	9.73±4.30	12.53±5.85	65.39±5.30	46.83±4.93
DER++	5	68.13±2.81	50.11±2.97	0.91±5.03	1.04±6.67	65.28±2.73	50.46±3.15
WACB	5	73.98±2.05	56.69±5.22	4.16±2.86	5.43±3.67	66.83±3.77	57.03±3.91
Freq	5	74.82±2.70	61.82±2.44	5.25±2.76	6.35±3.65	69.67±2.53	56.84±4.32
AA-CIL	5	74.24±1.29	58.26±1.85	11.12±2.70	14.75±3.70	62.84±3.46	55.66±3.36
TS-mem	5	72.65±2.03	55.62±1.58	10.53±3.63	13.73±4.66	63.19±4.66	51.44±2.16
AA-noAttMem	5	74.31±1.46	56.97±2.69	13.78±3.47	18.82±4.61	60.29±3.16	56.67±3.34
Replay-BS (post hoc)	5	75.05±1.01	59.32±2.85	11.44±2.53	15.15±3.51	62.48±4.64	57.05±3.33
AA-CIL-KD	5	76.39±0.26	60.82±0.73	0.98±0.76	1.27±1.17	71.67±0.66	61.43±2.25
AA-CIL-ISO	5	58.29±0.60	36.50±0.58	0.06±0.14	-0.06±0.05	72.41±0.24	31.77±1.09
AA-CIL-WA	5	76.17±0.67	59.33±1.09	2.37±0.59	3.02±0.66	72.41±0.24	59.85±1.87
AA-CIL-CBF	5	76.01±0.62	59.00±1.27	2.34±0.50	2.98±0.81	72.41±0.24	59.78±2.59
AA-CIL-NCM	5	76.16±0.27	59.98±0.68	2.35±1.00	3.09±1.33	71.67±0.66	60.05±1.81
FARQ (supp.)	5	74.57±2.57	58.22±4.10	4.70±5.00	5.99±6.40	67.45±2.77	58.86±5.79
FARQ-all (supp.)	5	73.21±3.72	54.47±12.14	14.21±10.57	18.86±14.08	63.08±2.48	49.57±11.97
Joint (offline)	5	74.62±0.69	58.65±2.20	3.45±0.81	4.51±0.93	66.63±1.80	59.37±2.90

Mean±sample SD across the available seeds; accuracies are percentages and forgetting is in percentage points. Mean incremental BAcc averages all seen-class stage BAcc values. Signed forgetting compares each old class's best pre-final recall with its final recall; the primary scope includes Normal. Smaller forgetting alone is not superiority: inspect final and incremental BAcc together. Groups with $n < 5$ are incomplete, descriptive only. Joint is offline; FARQ is supplementary; Replay+LA shares Replay's calibrated total τ and incurs no separate inference. The validation-selected retention protocol reports all five F variants; the main variant is fixed by development seeds 5/6, not by this test table. These post-hoc stagewise summaries do not change the original superiority protocol decision.

Table 28: UNSW: checkpoint-calibrated Class-IL performance at $\alpha = 2\%$.

Method	n	Mean inc. BAcc	Final BAcc	Forget (all old)	Forget (old att.)	Final ADR	Final MF1
Replay	5	52.29±0.55	25.15±0.89	5.09±1.02	5.82±1.17	38.52±3.77	26.27±0.85
Replay+LA	5	52.29±0.55	25.15±0.89	5.09±1.02	5.82±1.17	38.52±3.77	26.27±0.85
DER++	5	38.73±1.38	14.43±1.03	16.16±1.80	18.44±2.04	18.48±4.12	14.47±1.44
WACB	5	55.12±1.53	29.96±1.30	3.48±2.45	3.97±2.81	48.97±4.14	32.81±1.18
Freq	5	55.83±1.60	28.93±1.40	5.50±1.25	6.27±1.43	50.42±2.93	31.22±1.73
AA-CIL	5	69.65±0.49	55.05±1.06	20.04±1.16	22.92±1.32	69.09±0.93	41.18±0.28
TS-mem	5	48.14±3.67	24.97±7.34	9.73±5.85	11.11±6.69	24.79±14.93	23.47±6.33
AA-noAttMem	5	69.28±0.39	53.98±1.02	22.02±0.87	25.16±0.99	67.94±1.58	40.43±0.51
Replay-BS (post hoc)	5	62.16±0.24	38.01±1.19	4.87±2.44	5.54±2.78	66.14±0.74	39.99±1.57
AA-CIL-KD	5	63.75±1.11	41.80±1.93	2.31±1.07	2.62±1.20	70.53±0.50	42.08±1.93
AA-CIL-ISO	5	44.89±0.03	19.67±0.02	0.02±0.03	0.01±0.02	71.01±1.16	11.47±0.01
AA-CIL-WA	5	68.25±0.47	54.24±0.79	16.21±1.09	18.51±1.25	71.01±1.16	41.35±0.32
AA-CIL-CBF	5	68.73±0.49	54.69±0.70	15.32±0.56	17.49±0.64	71.01±1.16	40.86±0.38
AA-CIL-NCM	5	68.18±0.16	51.32±1.10	18.12±2.76	20.69±3.15	70.53±0.50	38.22±1.08
FARQ (supp.)	5	54.99±1.32	27.90±1.46	6.43±2.31	7.35±2.63	45.10±4.20	30.25±1.68
FARQ-all (supp.)	5	50.90±2.38	23.24±3.19	8.41±3.96	9.58±4.51	34.56±10.80	24.30±3.56
Joint (offline)	5	62.06±0.52	35.38±0.67	5.35±1.10	6.10±1.26	67.11±1.45	37.22±1.00

Mean±sample SD across the available seeds; accuracies are percentages and forgetting is in percentage points. Mean incremental BAcc averages all seen-class stage BAcc values. Signed forgetting compares each old class's best pre-final recall with its final recall; the primary scope includes Normal. Smaller forgetting alone is not superiority: inspect final and incremental BAcc together. Groups with $n < 5$ are incomplete, descriptive only. Joint is offline; FARQ is supplementary; Replay+LA shares Replay's calibrated total τ and incurs no separate inference. The validation-selected retention protocol reports all five F variants; the main variant is fixed by development seeds 5/6, not by this test table. These post-hoc stagewise summaries do not change the original superiority protocol decision.

tinct specification. This source check does not establish a causal explanation of the accuracy loss.The released means also place original AA-CIL below Replay-BS in final BAcc; this descriptive comparison does not establish significance. These are the earlier fixed-KD+WA results, whose failure is retained. The subsequent AR intervention removes gate distillation while retaining the family learner's DER-style replay and weight alignment; it is not the previously proposed redundant mask.

B.3 New-seed AR retention–plasticity follow-up

Tables 42–43 present the lead's final five-seed follow-up summary. AR does not meet final-accuracy superiority over Replay-BS at either operating point. Its lower forgetting is accompanied by a mean incremental BAcc cost of about 4–5 percentage points. This is a bounded RQ2 trade-off finding, not confirmation of an accuracy-improving method, causal proof of a gate mechanism, or closure of RQ1/RQ3 and the substantive-innovation requirement. Table 41 completes all 45 cells as a descriptive extension, not in the registered decision.

C Complete design-space tables

D Decision protocols and documented chronology

This appendix makes the decision rules self-contained. Times below are the repository's document-recording timestamps, not claims of external preregistration or evidence that a reader had inspected results. Unless stated otherwise, times are UTC+08:00 on 19 October 2026. Formal comparisons use the fixed paired seeds $\{1, 2, 3, 4, 42\}$, one-sided paired t tests at $p < 0.05$, and thresholds set only from the common permanent Normal training holdout at $\alpha = 2\%$. A partial group supplies descriptive means at its actual n, not a five-seed significance conclusion. A known necessary-condition

Table 29: CIC17: checkpoint-calibrated Class-IL performance at $\alpha = 2\%$.

Method	n	Mean inc. BAcc	Final BAcc	Forget (all old)	Forget (old att.)	Final ADR	Final MF1
Replay	5	74.30±1.20	52.32±4.55	6.99±7.30	8.38±8.76	78.89±5.63	51.52±4.02
Replay+LA	5	74.30±1.20	52.32±4.55	6.99±7.30	8.38±8.76	78.89±5.63	51.52±4.02
DER++	5	76.85±0.97	48.44±2.90	17.52±6.03	21.02±7.23	89.29±3.01	48.85±3.19
WACB	5	77.43±1.19	63.02±1.31	1.97±3.48	2.36±4.17	88.91±2.30	57.84±1.13
Freq	5	76.38±1.80	64.03±2.62	4.71±9.31	5.63±11.16	88.55±2.32	57.97±4.33
AA-CIL	5	93.95±0.46	87.54±0.85	8.99±1.57	10.77±1.88	83.66±2.83	63.15±1.14
TS-mem	5	66.04±1.55	40.34±2.01	26.70±3.69	32.03±4.42	49.51±3.90	34.63±2.25
AA-noAttMem	5	93.00±0.39	87.50±0.35	9.49±0.90	11.38±1.07	83.13±1.48	63.60±0.78
Replay-BS (post hoc)	5	90.82±1.83	79.26±3.61	3.58±10.18	4.29±12.22	92.45±1.15	69.10±0.92
AA-CIL-KD	5	87.59±1.07	79.20±4.08	-10.54±9.15	-12.66±10.99	92.10±2.84	65.94±2.49
FARQ (supp.)	5	75.72±1.79	57.54±5.40	1.20±7.71	1.43±9.25	85.40±5.48	55.13±2.26
FARQ-all (supp.)	5	71.58±3.86	50.30±9.58	8.28±15.62	9.93±18.73	74.49±10.03	48.47±6.84
Joint (offline)	5	93.83±0.80	78.68±3.54	13.14±2.71	15.75±3.26	98.19±1.94	68.43±1.30

Mean±sample SD across the available seeds; accuracies are percentages and forgetting is in percentage points. Mean incremental BAcc averages all seen-class stage BAcc values. Signed forgetting compares each old class's best pre-final recall with its final recall; the primary scope includes Normal. Smaller forgetting alone is not superiority: inspect final and incremental BAcc together. Groups with $n < 5$ are incomplete, descriptive only. Joint is offline; FARQ is supplementary; Replay+LA shares Replay's calibrated total τ and incurs no separate inference. The validation-selected retention protocol reports all five F variants; the main variant is fixed by development seeds 5/6, not by this test table. These post-hoc stagewise summaries do not change the original superiority protocol decision.

Table 30: NSL: AA-CIL old-attack recall-loss decomposition at $\alpha = 2\%$.

Old attack	n	Recall loss (pp)	Δ G miss (pp)	Δ F wrong (pp)
DoS	5	14.71±3.92	13.96±4.01	0.75±0.34
Probe	5	25.06±5.05	25.40±5.60	-0.34±1.31
R2L	5	4.49±9.23	-5.77±13.47	10.27±5.49

For each seed and class, the reference is its earliest highest-recall stage before the final task. Normal-miss and wrong-attack-family rates use all true-class rows as denominator; their signed changes add to recall loss. Means add, while component SDs do not. This observed confusion-matrix decomposition does not infer F behavior on G-rejected rows.

failure remains a failure when another comparison is incomplete.

D.1 Original superiority protocol

The revised original criteria were recorded at 14:31:21 (revision `aa16079`); the common holdout-calibration and separate gate/family update-budget clarification was recorded at 14:34:56 (`fe42938`). The comparator set contains Replay, DER++, WACB, Freq, Replay+LA, TS-mem and AA-noAttMem. The strongest comparator is chosen once per dataset and endpoint from the highest complete five-seed mean, never independently for individual seeds. The conjunction requires:

1. ADR significantly greater than the strongest comparator in at least two of NSL-KDD, UNSW-NB15 and CIC-IDS-2017.
2. Macro-F1 not significantly lower than its strongest comparator on any dataset; a one-sided lower-tail $p < 0.05$ identifies harm.
3. Attack-family macro recall not significantly lower than Replay, using attack classes with at least 20 test examples. All other per-class results remain reported.
4. On CIC-IDS-2017, AA-CIL Macro-F1 significantly greater than TS-mem Macro-F1.
5. Descriptive reporting of the offline Joint gap and training cost, without a numerical success threshold.

This protocol's superiority condition failed and is not replaced by the later external-only or retention comparisons.

D.2 Validation-selected retention protocol

The retention-constrained candidate-selection and decision rule was recorded at 18:42:22 (`e494c64`). Development uses only UNSW-NB15 seeds $\{5, 6\}$, permanent attack validation rows and the Normal calibration pool. Among configurations with final validation BAcc at least the original-AA development mean minus 3 percentage points, selection minimizes mean signed forgetting; this selects CBF. KD, ISO, WA, CBF and NCM are all reported on the formal seeds, including versions retained only as descriptive fallbacks after failing eligibility. The selected version must satisfy all three conditions on *both* UNSW-NB15 and NSL-KDD:

1. Signed all-old-class forgetting not significantly higher than the lowest-forgetting member of the external five: Replay, DER++, WACB, Freq and Replay+LA. The comparator minimizes the complete five-seed mean; an upper-tail $p < 0.05$ identifies harm.
2. Final BAcc significantly higher than exposure-matched Replay-BS (upper-tail $p < 0.05$).
3. ADR at the common calibration target not significantly lower than original AA-CIL (lower-tail $p < 0.05$ identifies harm).

Mean incremental BAcc, Macro-F1 and actual test FPR are mandatory descriptive outcomes. CIC-IDS-2017 is supplementary and cannot replace either required dataset. Failure of any necessary condition leaves the trade-off unresolved. The validation-selected CBF failure remains the outcome of this protocol.

Table 31: UNSW: AA-CIL old-attack recall-loss decomposition at $\alpha = 2\%$.

Old attack	n	Recall loss (pp)	Δ G miss (pp)	Δ F wrong (pp)
Generic	5	14.34±0.84	0.89±0.97	13.45±1.49
Exploits	5	43.78±3.40	4.31±1.91	39.46±5.06
Fuzzers	5	21.64±1.42	15.61±1.27	6.03±1.10
DoS	5	28.83±4.10	3.92±4.01	24.91±2.80
Reconnaissance	5	24.57±9.16	6.19±2.98	18.38±7.19
Analysis	5	25.56±8.35	26.67±7.93	-1.11±1.11
Backdoor	5	1.71±8.17	-8.29±6.88	10.00±3.64

For each seed and class, the reference is its earliest highest-recall stage before the final task. Normal-miss and wrong-attack-family rates use all true-class rows as denominator; their signed changes add to recall loss. Means add, while component SDs do not. This observed confusion-matrix decomposition does not infer F behavior on G-rejected rows.

Table 32: CIC17: AA-CIL old-attack recall-loss decomposition at $\alpha = 2\%$.

Old attack	n	Recall loss (pp)	Δ G miss (pp)	Δ F wrong (pp)
DoS	5	19.22±0.63	18.31±0.98	0.91±0.57
DDoS	5	3.70±1.33	3.56±1.36	0.14±0.18
Bruteforce	5	3.34±2.80	3.09±2.79	0.24±0.24
PortScan	5	25.30±8.44	25.10±8.37	0.21±0.15
WebAttack	5	2.30±2.94	0.22±0.16	2.08±2.92

For each seed and class, the reference is its earliest highest-recall stage before the final task. Normal-miss and wrong-attack-family rates use all true-class rows as denominator; their signed changes add to recall loss. Means add, while component SDs do not. This observed confusion-matrix decomposition does not infer F behavior on G-rejected rows.

D.3 Fixed-KD reassessment

The fixed-KD reassessment was recorded at 20:47:57 (`88d6355`), after the UNSW-NB15 formal outcomes had motivated reconsidering KD. It retains the frozen $\lambda_G = 0.5$ and family DER coefficients $\alpha = \beta = 1$, with no further tuning. It applies the same three forgetting, final-BAcc and ADR conditions immediately above, but requires *both NSL-KDD and CIC-IDS-2017*; UNSW-NB15 remains exploratory. Each comparison still requires its complete five-seed candidate and reference groups. Neither this reassessment nor any favorable descriptive KD outcome changes the validation-selected CBF decision.

NSL-KDD seeds 1 and 2 had completed before the reassessment document was recorded, and all five had completed before direct execution authorization at approximately 21:29. Result availability does not establish that the results were inspected, but it rules out describing this NSL evidence as blind confirmation. The reassessment's NSL final-BAcc superiority condition failed. The remaining CIC-IDS-2017 comparison is reported at its actual completeness and cannot reverse that necessary-condition failure. The original manuscript cutoff was later removed for completion of the remaining experiments; it does not redefine the criteria or turn partial trajectories into complete results.

D.4 Fixed-capacity retention protocol

The fixed-capacity extension was specified on 20 October 2026 before its development results. It preserves the earlier failed decisions and freezes each candidate using only UNSW/NSL validation seeds $\{5, 6\}$. Each candidate first requires signed validation forgetting no greater than the external-five minimum plus one percentage point on both datasets; among eligible configurations, mean validation final BAcc selects the configuration and main version. A BAcc tie within one point favors KD-weak/KD+WA/KD+CBF, then EMA-KD, then DUAL. An ineligible fallback is identified explicitly rather than recast as a successful selection.

The external CSE-CIC-IDS2018 assessment is separate from the development-dataset reports. Its frozen candidate must satisfy all four conditions: forgetting not significantly higher than the external-five minimum; final BAcc significantly higher than Replay-BS; mean final BAcc no more than two percentage points below original AA-CIL; and ADR not significantly below original AA-CIL. The two-point BAcc condition is a fixed mean-difference rule, not a significance or non-inferiority test. All 35 physical reference cells and five fixed-candidate cells completed before guarded release on 20 October 2026. The candidate and thresholds were not changed after release. Exact sign-flip tests enumerate all 32 paired-sign assignments as sensitivity checks; they do not replace the recorded paired-t decision rules. The resulting failure does not erase the earlier CBF or fixed-KD failures.

E Retention-extension paired comparisons

F Fixed-KD reassessment

G Fixed-capacity family counts

H Additional calibration targets

These results reuse the permanently held-out calibration source. They are sensitivity results, not a search over test thresholds. Actual test FPR accompanies each target, and an undefined per-class quantity remains undefined.

Table 33: Validation-selected retention protocol: frozen development configurations.

Version	λ_G	CBF epochs	CBF LR multiplier	Development role
AA-CIL-KD	0.5	–	–	ineligible fallback
AA-CIL-ISO	2	–	–	ineligible fallback
AA-CIL-WA	2	–	–	eligible alternative
AA-CIL-CBF	2	1	0.1	eligible main
AA-CIL-NCM	0.5	–	–	ineligible fallback

Selection uses UNSW training-validation seeds 5/6 only. DER weights are fixed at one. Ineligible versions are still reported on the five formal seeds, without converting fallback selection into eligibility.

Table 34: Retention variants at stagewise Normal-holdout calibration target $\alpha = 2\%$.

Data	Version	n	Mean inc. BAcc	Final BAcc	Forget (pp)	ADR	MF1	FPR
UNSW	AA-CIL-KD	5	63.75±1.11	41.80±1.93	2.31±1.07	70.53±0.50	42.08±1.93	2.10±0.18
UNSW	AA-CIL-ISO	5	44.89±0.03	19.67±0.02	0.02±0.03	71.01±1.16	11.47±0.01	2.04±0.08
UNSW	AA-CIL-WA	5	68.25±0.47	54.24±0.79	16.21±1.09	71.01±1.16	41.35±0.32	2.04±0.08
UNSW	AA-CIL-CBF	5	68.73±0.49	54.69±0.70	15.32±0.56	71.01±1.16	40.86±0.38	2.04±0.08
UNSW	AA-CIL-NCM	5	68.18±0.16	51.32±1.10	18.12±2.76	70.53±0.50	38.22±1.08	2.10±0.18
NSL	AA-CIL-KD	5	76.39±0.26	60.82±0.73	0.98±0.76	71.67±0.66	61.43±2.25	7.94±0.45
NSL	AA-CIL-ISO	5	58.29±0.60	36.50±0.58	0.06±0.14	72.41±0.24	31.77±1.09	8.40±0.37
NSL	AA-CIL-WA	5	76.17±0.67	59.33±1.09	2.37±0.59	72.41±0.24	59.85±1.87	8.40±0.37
NSL	AA-CIL-CBF	5	76.01±0.62	59.00±1.27	2.34±0.50	72.41±0.24	59.78±2.59	8.40±0.37
NSL	AA-CIL-NCM	5	76.16±0.27	59.98±0.68	2.35±1.00	71.67±0.66	60.05±1.81	7.94±0.45

Only complete seeds 1/2/3/4/42 receive means and sample SD; incomplete groups show n and dashes. All outcomes come from formal test trajectories, never development selection metrics. Signed forgetting includes old Normal; rates/accuracies are percentages. Eligibility is a development property, not formal success.

I Difficult-family performance

These selected families illustrate errors that binary attack detection alone cannot reveal. They are descriptive diagnostics, not a replacement endpoint or a selection rule for the main result; incomplete groups retain their actual seed counts.

J Registered safety contrasts and timing components

The timing components retain the measurement boundaries of their source records. In particular, a CPU analysis attempt and its nested inference or threshold-search components must not be summed twice.

Table 35: Fixed-capacity protocol: validation-only choices, not formal test results.

Candidate	λ_G	DER α/β	Mean val. BAcc	Mean val. forget	UNSW	NSL
KD-weak	0.5	0.25/0.5	73.82	5.79	no	no
KD+WA	0.5	1/1	73.18	2.09	yes	no
KD+CBF	0.5	1/1	73.40	6.40	no	no
EMA-KD	0.5	1/1	71.11	4.71	no	no
DUAL	0.5	1/1	73.26	7.81	no	no

Seeds 5/6 only. BAcc is in percent and signed forgetting in percentage points; means average the two development datasets. The last two columns indicate each dataset's forgetting constraint. Ceilings are 4.68 pp (UNSW) and -17.73 pp (NSL), each equal to its lowest external-five validation forgetting plus 1 pp. Negative values are not clipped. Rows with no eligible configuration are fixed descriptive fallbacks.

Table 36: Fixed-capacity candidates on development datasets at stagewise $\alpha = 2\%$.

Data	Candidate	n	Mean inc. BAcc	Final BAcc	Forget (pp)	ADR	MF1	FPR
UNSW	KD-weak	5	70.08±0.49	56.73±1.57	11.75±1.22	70.53±0.50	43.39±0.69	2.10±0.18
UNSW	KD+WA	5	68.63±0.70	55.17±1.26	5.70±0.97	70.53±0.50	44.77±0.34	2.10±0.18
UNSW	KD+CBF	5	70.15±0.45	56.66±1.51	11.68±1.73	70.53±0.50	43.53±0.71	2.10±0.18
UNSW	EMA-KD	5	69.80±0.51	56.50±1.00	8.51±1.22	70.53±0.50	43.74±0.71	2.10±0.18
UNSW	DUAL	5	69.93±0.38	56.63±1.04	14.77±2.35	70.53±0.50	42.29±0.33	2.10±0.18
NSL	KD-weak	5	76.81±0.49	61.36±0.96	1.52±0.63	71.67±0.66	61.38±1.61	7.94±0.45
NSL	KD+WA	5	76.77±0.41	61.19±0.59	1.53±0.85	71.67±0.66	61.58±1.21	7.94±0.45
NSL	KD+CBF	5	76.83±0.30	61.30±1.35	1.65±1.11	71.67±0.66	60.62±1.60	7.94±0.45
NSL	EMA-KD	5	76.15±0.55	59.97±0.63	2.88±1.15	71.67±0.66	57.01±1.25	7.94±0.45
NSL	DUAL	5	76.01±0.42	58.44±1.91	4.23±2.03	71.67±0.66	59.04±2.20	7.94±0.45
CIC17	KD+WA	5	87.58±1.08	79.26±4.12	-10.54±9.16	92.10±2.84	65.23±1.89	1.93±0.04

Mean±sample SD over complete seeds 1/2/3/4/42. UNSW, NSL and CIC17 were used for development; these five-seed results are not blind confirmation. CIC17 was not used in the frozen UNSW/NSL validation-only selection; only its selected candidate was added. Only complete groups are displayed. No test outcome changes the fixed selection or fallback eligibility.

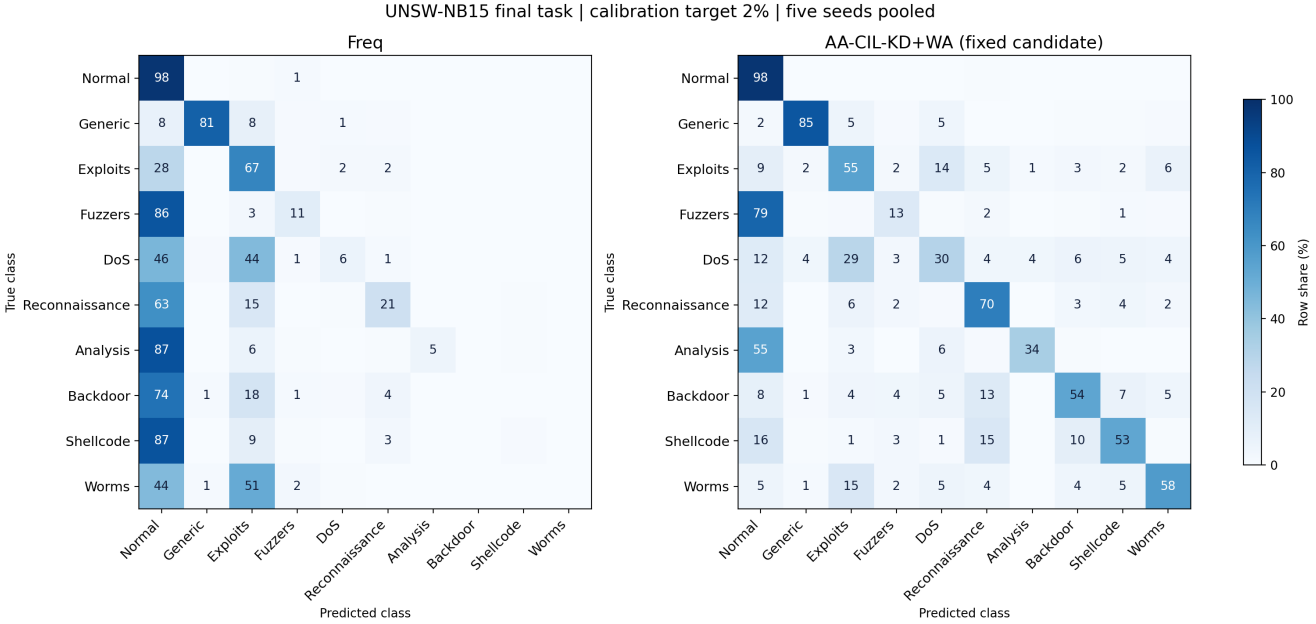


Figure 5: UNSW-NB15 final-task confusion at the $\alpha = 2\%$ calibration target: Freq and the fixed AA-CIL-KD+WA candidate. Saved confusion counts for seeds 1, 2, 3, 4 and 42 are summed before row normalization; each seed evaluates the same 30,248 test rows, so pooling does not create independent observations. Cells show percentages rounded to integers when at least 1%; blank labels are below 1%, not necessarily zero. The target concerns calibration, not an exact 2% test FPR. This is descriptive development evidence, not a causal ablation or confirmation of a final method.

Table 37: CIC-IDS-2017 development results at saved stagewise $\alpha = 2\%$: five-seed means.

Method	AIA	Final BAcc	Signed forget.	ADR	Macro-F1
AA-CIL-KD+WA	87.6	79.3	-10.5	92.1	65.2
AA-CIL-KD	87.6	79.2	-10.5	92.1	65.9
AA-CIL	93.9	87.5	9.0	83.7	63.1
Replay-BS	90.8	79.3	3.6	92.4	69.1
WACB	77.4	63.0	2.0	88.9	57.8

Rates are percentages; signed forgetting is in percentage points and is not clipped. Seeds are 1, 2, 3, 4, 42. AA-CIL-KD+WA is the fixed validation-selected candidate; CIC17 was not used for frozen validation selection. These are descriptive development results, not CSE18 confirmation. Sample SDs and per-seed values remain in the accompanying result artifacts.

Table 38: CSE18: compact retention at stagewise $\alpha = 2\%$.

Method	n	Mean inc. BAcc	Final BAcc	Forget (pp)	ADR	MF1	FPR
KD+WA*	5	82.64±1.38	72.69±1.96	5.27±0.98	95.44±0.11	66.76±0.20	1.95±0.06
Replay	5	53.50±8.74	26.31±14.28	30.85±14.76	12.90±25.90	22.68±17.21	2.04±0.04
DER++	5	48.81±7.30	14.98±0.44	34.55±13.23	1.50±0.94	14.27±0.83	2.04±0.01
WACB	5	60.51±6.30	51.24±20.54	12.58±29.74	54.16±33.69	48.66±22.33	2.01±0.04
Freq	5	71.93±0.54	72.35±2.46	5.79±2.54	92.17±3.19	64.24±2.84	2.00±0.04
Replay-BS	5	86.58±0.96	79.74±2.03	5.86±1.56	95.04±0.31	66.63±0.32	1.98±0.05
AA-CIL	5	85.14±0.48	78.53±1.87	3.86±2.80	94.84±0.48	65.93±0.68	1.98±0.02
AA-CIL-KD	5	84.31±0.71	73.97±1.37	0.58±1.16	95.48±0.10	71.17±0.23	1.95±0.06
Replay+LA	5	53.50±8.74	26.31±14.28	30.85±14.76	12.90±25.90	22.68±17.21	2.04±0.04

Mean±sample SD over all five fixed formal seeds; signed forgetting is in percentage points. $\star$ identifies the validation-selected main, never a test-selected winner. External data are released only after all 35 reference cells finish. Replay+LA aliases calibrated Replay and adds no training.

Table 39: Compact family parameter and component counts.

Data	Method	Training-resident	Optimizer	Inference	Extra components
CSE18	KD+WA	27398	27398	27398	0

F only; the common G is separate. Counts are fixed across stages. EMA training-resident count includes its frozen teacher; inference uses only EMA. DUAL is one F with two fixed encoders and a joint head. Sample, calibration and transient storage and separate gate/family costs remain in the generated JSON.

Table 40: External-dataset necessary tests for the fixed validation-selected main.

Criterion	Reference	n	Δ (pp)	t p	Sign-flip p	Decision
1	Freq	5	-0.52±2.12	0.693	0.7188	pass
2	Replay-BS	5	-7.06±1.87	0.9995	1	fail
3	AA-CIL	5	-5.85±2.25	not used	not used	fail
4	AA-CIL	5	0.60±0.42	0.9838	1	pass

1: forgetting harm is $p_> < .05$ against the lowest-mean external-five member. 2: final BAcc superiority over Replay-BS requires $p_> < .05$. 3: mean final-BAcc difference from original AA must be at least -2pp; no p-test is used. 4: ADR harm is $p_< < .05$ against original AA. No detected harm is not equivalence. Exact sign-flip p-values enumerate all 32 assignments as a sensitivity check. Overall numerical decision: fail.

Table 41: All 45 CSE18 cells: nine methods, seeds 5–9, calibration target 2%. Descriptive extension: not in the registered decision.

Method	Mean inc. BAcc	Final BAcc	Forget (pp)	ADR	Test FPR
AR	81.67 ± 4.35	77.44 ± 2.89	-0.24 ± 6.32	95.04 ± 0.18	1.97 ± 0.07
AA-CIL	84.73 ± 0.56	77.36 ± 2.52	4.08 ± 3.96	95.04 ± 0.18	1.97 ± 0.07
Replay-BS	86.21 ± 0.48	77.52 ± 3.84	8.26 ± 5.50	94.26 ± 1.63	1.99 ± 0.06
Freq	69.86 ± 6.07	61.46 ± 17.08	14.25 ± 10.11	75.81 ± 25.60	2.00 ± 0.03
WACB	65.54 ± 11.16	50.12 ± 26.27	19.93 ± 16.55	50.75 ± 44.77	2.03 ± 0.04
KD+WA	78.94 ± 4.62	70.44 ± 2.33	3.31 ± 4.25	95.23 ± 0.41	1.95 ± 0.06
Replay	55.94 ± 3.44	21.37 ± 5.40	40.88 ± 5.21	7.06 ± 10.95	2.04 ± 0.03
DER++	53.21 ± 5.35	19.70 ± 10.62	36.36 ± 19.41	13.37 ± 26.88	2.05 ± 0.03
AKD	72.53 ± 9.67	56.73 ± 23.43	10.02 ± 19.15	74.99 ± 35.32	2.00 ± 0.03

Final BAcc by seed (percent; all five values, no selected seeds)

Method	Seed 5	Seed 6	Seed 7	Seed 8	Seed 9
AR	76.27	80.88	78.81	78.05	73.19
AA-CIL	76.58	74.35	78.99	80.75	76.14
Replay-BS	71.49	81.94	79.16	77.97	77.03
Freq	31.89	61.72	73.32	70.45	69.92
WACB	21.35	21.81	64.33	74.00	69.10
KD+WA	71.82	72.67	71.49	66.89	69.35
Replay	15.13	21.23	21.41	29.92	19.16
DER++	14.86	15.08	38.69	15.61	14.29
AKD	65.71	17.86	75.06	52.47	72.57

Upper panel: equal-weight means ± sample SD across five seeds, not confidence intervals. Values are percentages except signed forgetting in percentage points, including Normal; negative values are not clipped. The 2% target is a calibration setting, not a guaranteed test FPR. Source: published Section 64 single-seed releases (AR, AA-CIL, Replay-BS, Freq, WACB), Section 74 descriptive releases (KD+WA, Replay, DER++), and Section 82 descriptive releases (AKD). Aggregation uses saved scalar metrics only; AKD ADR/FPR aliases are checked against the saved final operating point. No training, inference, confusion-matrix recomputation, or new significance test is involved. The 40 original cells and five separately appended AKD cells are counted once; Replay-LA is not an additional physical method. This nine-method extension is not in the registered decision: the original five-method decision and its failures remain unchanged, and the four additional methods are descriptive only. These are new seeds on the same exposed CSE dataset, not independent-dataset confirmation. Complete-seed interim releases do not authorize tuning, early stopping, or reselection.

Table 42: CSE18, paired seeds 5–9 ($n = 5$): retention–plasticity trade-off. Blind seeds 5–9; externals evaluated from saved checkpoints, no retraining (1% panel).

Target	Method	Mean inc. BAcc	Final BAcc	Forget (pp)	ADR	Test FPR
2%	AR	81.7	77.4	−0.2	95.0	1.97
	AA-CIL	84.7	77.4	4.1	95.0	1.97
	Replay-BS	86.2	77.5	8.3	94.3	1.99
	Freq	69.9	61.5	14.2	75.8	2.00
	WACB	65.5	50.1	19.9	50.7	2.03
1%	AR	80.9	76.3	−0.8	94.5	1.00
	AA-CIL	83.7	76.2	3.8	94.5	1.00
	Replay-BS	84.7	74.4	8.2	93.3	1.01
	Freq	63.9	58.3	11.4	68.2	1.00
	WACB	50.2	28.7	15.8	18.8	1.03

Values are percentages except signed forgetting in percentage points; negative forgetting is retained, not clipped. Rounded means are transcribed from the lead's recomputation recorded in work-order Sections 71–72, except the completed 2% ADR and test-FPR columns, which are arithmetic means of the five saved final values per method in the published Section 64 seed releases. No confusion matrices or statistical tests were recomputed for this completion. The 2% panel derives from the 25 published decision cells; the 1% panel combines saved AR/AA/BS operating points with Freq/WACB checkpoint evaluation. No model was retrained for the 1% panel. "Blind seeds" describes the initially withheld seed results, not an independent dataset: the design followed unsealing of the earlier CSE round, uses the same exposed dataset, and complete single-seed interim releases preceded this final summary. No early stopping, tuning, or criterion changes followed those releases.

Table 43: CSE18 lead-reported paired tests at 1% ($n = 5$). Differences are AR minus reference, in percentage points.

Reference	Endpoint (alternative)	Difference	One-sided t p	Exact sign-flip p
Freq	Final BAcc (>)	18.0	.045	.031
Freq	Mean inc. BAcc (>)	17.0	.005	.031
WACB	Final BAcc (>)	47.6	.001	.031
WACB	Mean inc. BAcc (>)	30.7	.003	.031
Replay-BS	Final BAcc (>)	1.9	.059	.062

Rounded values from work-order Sections 71–72; paired t has four degrees of freedom, and sign-flip enumerates all 32 sign assignments. These are reported tests, not new tests executed during typesetting. At 1%, the lead reports AR forgetting differences of −12.1pp versus Freq and −16.5pp versus WACB; harm is not significant against the lowest-forgetting external (Freq). Thus the external half of H-low passes for AR, but the Replay-BS superiority half fails, so the conjunction is not confirmed. This is not an equivalence finding. At 2%, the registered final-BAcc superiority criterion also fails (AR minus BS -0.08 ± 3.11pp; reported one-sided $p \geq .52$), while the other three criteria pass. No new round or selection is implied. AR has the lowest mean forgetting among these five available decision methods, not among unreported variants.

Table 44: Design-space study; no configuration met the pre-registered K1. Screening seeds 1–2 only. Values are percentages (forgetting in percentage points). Score averages both final decision views over both datasets and seeds. D: default final balanced accuracy; L: final balanced accuracy at a 1% calibration FPR target; F: signed old-class forgetting in the latter view, including Normal. * denotes selected configurations; R denotes the reference. These screening means are not confirmation evidence.

Configuration	Score	UNSW D	UNSW L	UNSW F	CIC D	CIC L	CIC F
a2b2c1d3e1*	75.13	62.92	53.34	8.46	94.39	89.87	3.17
a2b2c1d1e1*	74.55	61.80	52.24	9.93	94.34	89.82	3.25
a2b2c1d1e2	74.53	61.73	52.47	17.98	94.20	89.71	3.39
a2b2c1d2e1*	74.40	61.34	51.97	6.90	94.40	89.88	3.18
a2b2c2d1e2	74.39	61.34	52.62	19.79	93.87	89.74	2.98
a2b1c2d1e2	74.39	61.87	51.58	20.98	94.25	89.85	2.88
a2b1c2d1e1	74.38	62.14	51.03	12.40	94.38	89.99	2.65
a2b1c1d1e2	74.32	63.46	52.84	18.36	94.25	86.75	7.58
a2b2c2d3e1	74.09	60.58	51.92	7.76	94.01	89.83	2.93
a2b2c2d2e1	74.04	60.32	52.02	6.95	93.99	89.83	2.94
a2b2c2d1e1	74.01	60.48	51.79	11.03	93.97	89.79	2.94
a2b1c1d3e1	73.75	62.40	51.41	7.38	94.35	86.83	7.46
a2b1c2d3e1	73.74	60.53	49.93	6.76	94.45	90.05	2.57
a2b1c1d2e1	73.74	62.34	51.41	5.93	94.36	86.84	7.45
a2b1c2d2e1	73.71	60.34	49.99	6.24	94.45	90.04	2.57
a2b1c1d1e1	73.68	62.37	51.27	10.03	94.29	86.79	7.52
a1b2c2d2e1	73.07	60.82	52.94	7.39	92.05	86.47	-5.54
a1b2c2d3e1	72.85	60.38	52.62	8.39	91.99	86.42	-5.48
a1b1c1d3e1	72.58	62.91	52.83	7.73	91.49	83.10	-3.46
a1b2c2d1e1	72.58	59.75	52.05	11.74	92.05	86.47	-5.51
a1b2c1d3e1	72.50	62.50	52.28	8.56	91.54	83.67	-3.68
a1b1c2d2e1	72.45	62.80	52.22	8.57	91.66	83.11	-3.34
a1b1c1d1e2	72.33	62.47	52.07	19.40	91.59	83.20	-3.58
a1b1c1d1e1	72.31	62.25	52.34	10.08	91.51	83.13	-3.48
a1b2c1d1e2	72.22	61.97	51.58	18.96	91.61	83.74	-3.69
a1b1c1d2e1 R	72.22	61.93	52.30	7.02	91.52	83.13	-3.48
a1b1c2d3e1	72.21	62.34	51.84	9.21	91.60	83.06	-3.27
a1b2c1d1e1	72.19	61.80	51.72	10.43	91.57	83.69	-3.63
a1b2c2d1e2	72.19	58.66	51.09	19.26	92.30	86.71	-5.83
a1b2c1d2e1	72.17	61.60	51.83	7.65	91.57	83.68	-3.70
a1b1c2d1e1	71.98	61.81	51.33	12.56	91.67	83.11	-3.31
a1b1c2d1e2	71.69	60.77	50.70	19.68	91.93	83.36	-3.66
a2b2c2d3e2	71.46	54.87	47.38	6.20	93.86	89.75	3.02
a2b1c2d3e2	71.33	55.27	45.85	6.45	94.30	89.89	2.81
a2b2c1d2e2	71.20	54.40	46.37	5.30	94.27	89.77	3.31
a2b2c1d3e2	70.99	53.99	45.97	6.47	94.25	89.76	3.31
a2b2c2d2e2	70.53	52.79	45.62	6.02	93.92	89.76	2.98
a2b1c2d2e2	70.35	52.95	44.62	6.00	94.10	89.72	3.02
a2b1c1d3e2	70.30	54.63	45.49	4.58	94.29	86.78	7.48
a1b2c2d3e2	70.20	54.72	47.68	6.65	92.00	86.41	-5.48
a2b1c1d2e2	69.84	53.80	44.58	4.54	94.24	86.73	7.55
a1b1c2d3e2	69.63	56.58	47.20	6.35	91.63	83.10	-3.31
a1b2c2d2e2	69.31	52.91	45.89	6.72	92.01	86.43	-5.49
a1b2c1d3e2	68.89	54.85	45.70	4.44	91.40	83.59	-3.69
a1b1c1d3e2	68.88	55.05	46.07	3.91	91.38	83.00	-3.55
a1b1c2d2e2	68.72	54.58	45.57	7.17	91.63	83.10	-3.30
a1b2c1d2e2	68.22	52.95	44.68	4.73	91.55	83.70	-3.70
a1b1c1d2e2	68.19	53.11	45.03	4.50	91.51	83.12	-3.56

Table 45: Confirmation-stage descriptive results, seeds 10–14, mean ± sample standard deviation. All columns use the 1% calibration FPR target, not default decisions or guaranteed test FPR. All three selected configurations failed K1 on both datasets. K2 passing means no significant worsening under the original test; it does not establish equivalence or non-inferiority. K3 passed under the original criterion. No configuration was confirmed.

Dataset	Configuration	Mean incremental BAcc	Final BAcc	Signed forgetting
UNSW	a2b2c1d3e1	66.2 ± 0.7	51.2 ± 1.5	8.8 ± 1.2
UNSW	a2b2c1d1e1	66.9 ± 0.4	51.2 ± 1.2	11.6 ± 1.4
UNSW	a2b2c1d2e1	66.1 ± 0.7	50.8 ± 1.0	8.1 ± 1.0
UNSW	ar	66.3 ± 0.8	51.0 ± 1.6	9.5 ± 1.3
UNSW	ar_joint	63.6 ± 0.6	45.7 ± 1.3	24.0 ± 2.9
UNSW	replay_bs	59.3 ± 0.7	33.2 ± 1.9	5.0 ± 1.8
UNSW	freq	52.5 ± 1.6	27.5 ± 1.2	2.4 ± 1.7
UNSW	wacb	50.5 ± 3.0	27.2 ± 2.4	1.4 ± 2.5
UNSW	er_bal_tabt	69.8 ± 0.5	56.3 ± 1.6	25.0 ± 1.0
CIC	a2b2c1d3e1	96.9 ± 0.2	90.0 ± 0.4	2.9 ± 0.4
CIC	a2b2c1d1e1	96.9 ± 0.3	89.9 ± 0.5	2.9 ± 0.5
CIC	a2b2c1d2e1	96.9 ± 0.2	90.0 ± 0.5	2.9 ± 0.4
CIC	ar	86.3 ± 1.1	81.6 ± 3.0	-3.1 ± 1.5
CIC	ar_joint	85.8 ± 0.6	79.3 ± 4.4	0.4 ± 7.2
CIC	replay_bs	80.3 ± 2.4	57.4 ± 6.0	11.4 ± 5.1
CIC	freq	69.3 ± 3.1	48.1 ± 4.7	8.5 ± 5.6
CIC	wacb	68.4 ± 1.5	45.9 ± 2.2	5.1 ± 4.3
CIC	er_bal_tabt	96.7 ± 0.6	92.7 ± 2.0	2.7 ± 0.3

Table 46: Original registered superiority test and separately identified external-only comparison. The latter does not replace the former.

Data	Comparison set	Fixed mean winner	Δ ADR (pp)	$p_>$	d_z	Dataset test
NSL	registered	Freq	-6.829±3.350	0.99482	-2.039	Fail
NSL	external	Freq	-6.829±3.350	0.99482	-2.039	Fail
UNSW	registered	AA-noAttMem	1.154±1.450	0.07480	0.796	Fail
UNSW	external	Freq	18.671±3.277	0.00011	5.697	Pass
CIC17	registered	DER++	-5.632±4.149	0.98070	-1.357	Fail
CIC17	external	DER++	-5.632±4.149	0.98070	-1.357	Fail

Original superiority protocol, criteria 1–5: 1: FAIL, 2: PASS, 3: PASS, 4: PASS, 5: PASS. Overall decision: FAIL, because NSL and UNSW cannot contribute the required two significant wins. The comparator set includes TS-mem and AA-noAttMem.

Table 47: Necessary comparisons under the validation-selected retention protocol.

Data	Criterion	Reference	n	Δ (pp)	One-sided p	Decision
UNSW	1	WACB	5	11.84±2.91	0.00041	fail
UNSW	2	Replay-BS (post hoc)	5	16.67±1.26	3.868×10^{-6}	pass
UNSW	3	AA-CIL	5	1.92±1.29	0.98539	pass
NSL	1	DER++	5	1.42±5.29	0.28991	pass
NSL	2	Replay-BS (post hoc)	5	-0.33±2.73	0.59933	fail
NSL	3	AA-CIL	5	9.57±3.63	0.99794	pass

Candidate minus reference; exact five paired formal seeds. 1: all-old forgetting, $p_> < .05$ means harm/failure against the lowest-forgetting external baseline. 2: final BAcc, $p_> < .05$ is required against Replay-BS. 3: ADR, $p_< < .05$ means harm/failure against original AA-CIL. Only useful complete comparisons are shown; absent comparisons remain pending and are not passed. Both UNSW and NSL are required. No detected harm is not equivalence. Unadjusted tests.

Table 48: Completed descriptive AA-CIL-ISO minus AA-CIL-KD comparisons.

Data	Metric	n	Δ (pp)	$p_<$	$p_>$	p_{2s}
UNSW	Forget (all)	5	-2.29±1.03	0.00385	0.99615	0.00770
UNSW	Mean inc. BAcc	5	-18.86±1.10	1.365×10^{-6}	$1 - 1.365 \times 10^{-6}$	2.731×10^{-6}
NSL	Forget (all)	5	-0.93±0.89	0.03981	0.96019	0.07963
NSL	Mean inc. BAcc	5	-18.10±0.55	1.052×10^{-7}	$1 - 1.052 \times 10^{-7}$	2.104×10^{-7}

Exact five paired formal seeds; mean±sample SD of differences in percentage points. These unadjusted comparisons neither select the main version nor add a success requirement. Undefined tests remain –; no seed is discarded. The original superiority and earlier retention decisions are unchanged.

Table 49: Fixed-KD reassessment at stagewise Normal-holdout calibration target $\alpha = 2\%$.

Data	n	Mean inc. BAcc	Final BAcc	Forget (pp)	ADR	MF1	FPR
UNSW	5	63.75±1.11	41.80±1.93	2.31±1.07	70.53±0.50	42.08±1.93	2.10±0.18
NSL	5	76.39±0.26	60.82±0.73	0.98±0.76	71.67±0.66	61.43±2.25	7.94±0.45
CIC17	5	87.59±1.07	79.20±4.08	-10.54±9.15	92.10±2.84	65.94±2.49	1.93±0.04

Mean±sample SD over completed trajectories at the stated n; $n = 1$ has undefined SD (–). Incomplete groups are descriptive only; significance requires all five formal seeds. UNSW is exploratory. NSL results were available before direct authorization, so NSL is not blind confirmation. KD uses fixed $\lambda_G = .5$ and DER $\alpha = \beta = 1$. The validation-selected CBF and original superiority decisions remain failed.

Table 50: Fixed-KD reassessment: candidate minus reference.

Data	Criterion	Reference	n	Δ (pp)	One-sided p	Decision
UNSW	1	WACB	5	-1.17±3.41	0.7563	pass
UNSW	2	Replay-BS (post hoc)	5	3.78±1.43	0.002032	pass
UNSW	3	AA-CIL	5	1.44±0.85	0.9905	pass
NSL	1	DER++	5	0.07±4.87	0.4876	pass
NSL	2	Replay-BS (post hoc)	5	1.49±2.58	0.1329	fail
NSL	3	AA-CIL	5	8.84±2.94	0.9987	pass
CIC17	1	WACB	5	-12.51±10.48	0.9721	pass
CIC17	2	Replay-BS (post hoc)	5	-0.07±5.80	0.5095	fail
CIC17	3	AA-CIL	5	8.44±2.61	0.999	pass

1: forgetting harm, $p_> < .05$ means failure against the lowest-forgetting external-five member. 2: final BAcc superiority over Replay-BS requires $p_> < .05$. 3: ADR harm relative to original AA means $p_< < .05$. NSL and CIC17 are both required; UNSW does not replace either. Unadjusted tests; no detected harm is not equivalence. Required numerical verdict: fail.

Table 51: Fixed family-model capacity of the selected per-candidate configurations.

Data	Candidate	Training-resident	Optimizer	Inference	Extra components
UNSW	KD-weak	26505	26505	26505	0
NSL	KD-weak	32004	32004	32004	0
UNSW	KD+WA	26505	26505	26505	0
NSL	KD+WA	32004	32004	32004	0
UNSW	KD+CBF	26505	26505	26505	0
NSL	KD+CBF	32004	32004	32004	0
UNSW	EMA-KD	53010	26505	26505	1
NSL	EMA-KD	64008	32004	32004	1
UNSW	DUAL	53001	53001	53001	1
NSL	DUAL	64004	64004	64004	1
CIC17	KD+WA	27399	27399	27399	0

F only; the common G and transient previous-task gate teacher are separate. Counts come from saved development architectures and remain fixed across tasks. EMA training-resident count includes its slow copy; only EMA is used at inference. DUAL uses two fixed encoders and one joint head within a single F.

Table 52: Additional calibration targets: NSL-KDD. Rates in percent; mean ± sample SD.

Method	α	n	Test FPR	Detection	MF1
Replay	1	5	2.60±1.82	56.54±8.96	36.14±8.07
Replay	5	5	5.92±2.76	76.73±3.63	59.43±3.60
DER++	1	5	1.45±0.35	58.01±6.83	42.04±7.13
DER++	5	5	4.44±0.74	73.86±3.16	60.62±2.98
WACB	1	5	2.74±0.25	61.74±4.42	48.34±5.05
WACB	5	5	5.31±2.02	75.34±4.53	57.78±1.61
Freq	1	5	4.54±2.44	66.24±3.69	57.79±3.89
Freq	5	5	6.34±2.74	75.94±4.01	55.34±2.40
Replay+LA	1	5	2.60±1.82	56.54±8.96	36.14±8.07
Replay+LA	5	5	5.92±2.76	76.73±3.63	59.43±3.60
TS-mem	1	5	2.32±1.85	51.03±6.90	45.92±4.77
TS-mem	5	5	6.10±2.69	73.74±4.61	55.52±2.29
AA-noAttMem	1	5	3.59±1.57	57.20±1.49	56.68±3.42
AA-noAttMem	5	5	6.18±2.36	67.14±2.65	56.35±2.93
AA-CIL	1	5	4.89±1.91	58.93±2.19	53.54±2.62
AA-CIL	5	5	7.14±1.75	67.83±3.44	57.44±1.81
Replay-BS (post hoc)	1	5	3.16±1.58	59.01±4.04	58.43±2.41
Replay-BS (post hoc)	5	5	4.36±2.25	66.84±4.27	55.84±2.69
Joint (offline)	1	5	6.30±1.23	64.23±1.29	57.26±2.41
Joint (offline)	5	5	7.61±0.85	71.12±2.70	57.37±1.03

Table 53: Additional calibration targets: UNSW-NB15. Rates in percent; mean ± sample SD.

Method	α	n	Test FPR	Detection	MF1
Replay	1	5	0.86±0.18	28.97±4.87	23.40±0.90
Replay	5	5	4.80±0.19	56.73±2.53	32.85±1.80
DER++	1	5	0.99±0.14	9.69±5.65	11.23±2.31
DER++	5	5	4.87±0.42	40.57±6.65	26.89±5.22
WACB	1	5	0.95±0.05	39.02±5.88	27.80±2.02
WACB	5	5	4.90±0.35	65.89±1.88	42.17±0.72
Freq	1	5	0.98±0.17	41.24±3.41	26.95±2.00
Freq	5	5	5.00±0.33	65.64±1.99	41.21±1.82
Replay+LA	1	5	0.86±0.18	28.97±4.87	23.40±0.90
Replay+LA	5	5	4.80±0.19	56.73±2.53	32.85±1.80
TS-mem	1	5	0.96±0.09	11.78±10.64	17.15±6.58
TS-mem	5	5	5.22±0.35	55.95±5.94	34.85±1.90
AA-noAttMem	1	5	1.02±0.12	61.27±3.58	39.15±0.80
AA-noAttMem	5	5	5.07±0.30	76.24±0.75	40.96±0.34
AA-CIL	1	5	1.04±0.09	63.51±1.40	40.22±0.39
AA-CIL	5	5	5.05±0.17	76.83±0.62	41.67±0.50
Replay-BS (post hoc)	1	5	1.08±0.12	59.65±0.93	36.02±1.56
Replay-BS (post hoc)	5	5	5.13±0.29	75.85±1.11	47.54±1.61
Joint (offline)	1	5	1.00±0.10	61.09±1.83	33.36±0.46
Joint (offline)	5	5	5.08±0.33	77.39±1.51	41.60±0.74

Table 54: Additional calibration targets: CIC-IDS-2017. Rates in percent; mean ± sample SD.

Method	α	n	Test FPR	Detection	MF1
Replay	1	5	0.98±0.04	67.05±8.32	44.49±6.18
Replay	5	5	4.87±0.06	94.85±1.90	57.05±1.92
DER++	1	5	0.96±0.03	72.99±7.31	42.92±2.54
DER++	5	5	4.86±0.03	98.28±0.27	62.87±0.65
WACB	1	5	0.95±0.04	79.18±2.75	50.23±4.04
WACB	5	5	4.85±0.02	97.21±0.60	61.05±0.98
Freq	1	5	0.97±0.04	81.16±2.48	50.00±4.07
Freq	5	5	4.83±0.03	96.46±0.76	59.82±1.49
Replay+LA	1	5	0.98±0.04	67.05±8.32	44.49±6.18
Replay+LA	5	5	4.87±0.06	94.85±1.90	57.05±1.92
TS-mem	1	5	0.94±0.01	40.22±6.08	30.28±1.91
TS-mem	5	5	5.07±0.45	75.51±6.27	51.01±3.59
AA-noAttMem	1	5	0.98±0.01	78.32±1.40	64.10±1.21
AA-noAttMem	5	5	4.88±0.03	94.57±2.17	62.32±1.02
AA-CIL	1	5	0.96±0.01	78.23±2.71	64.35±1.36
AA-CIL	5	5	4.89±0.03	94.52±2.06	62.02±0.97
Replay-BS (post hoc)	1	4	0.95±0.03	87.25±1.74	59.51±6.41
Replay-BS (post hoc)	5	4	4.86±0.03	96.50±1.67	62.32±1.35
Joint (offline)	1	5	0.94±0.05	94.24±2.98	63.48±5.19
Joint (offline)	5	5	4.85±0.02	99.87±0.12	63.96±1.39

Table 55: Selected low-support or weak-family results at $\alpha = 2\%$, to expose failure modes hidden by binary detection. All classes remain in Macro-F1 and BAcc; only the attack guard excludes support below 20.

Data	Method	Class	n	Support	n_P	Precision	Recall	F1
NSL	AA-CIL	R2L	5	2885	5	79.21±20.34	19.34±5.56	30.98±8.50
NSL	AA-CIL	U2R	5	67	5	14.36±15.83	46.27±2.36	19.07±14.54
NSL	TS-mem	R2L	5	2885	5	49.71±22.71	9.74±4.42	16.03±6.95
NSL	TS-mem	U2R	5	67	5	6.68±2.86	40.30±8.31	11.24±4.03
NSL	AA-noAttMem	R2L	5	2885	5	79.36±13.06	15.45±4.49	25.71±6.83
NSL	AA-noAttMem	U2R	5	67	5	26.25±18.85	43.28±6.76	29.57±17.65
NSL	Replay-BS (post hoc)	R2L	5	2885	5	95.22±2.37	13.72±4.07	23.82±6.15
NSL	Replay-BS (post hoc)	U2R	5	67	5	25.86±22.55	47.76±8.24	27.30±16.01
UNSW	AA-CIL	Fuzzers	5	3750	5	69.58±4.05	9.67±1.14	16.94±1.67
UNSW	AA-CIL	Worms	5	32	5	3.52±0.45	63.75±6.09	6.68±0.83
UNSW	TS-mem	Fuzzers	5	3750	5	47.54±14.46	3.39±2.15	6.28±3.88
UNSW	TS-mem	Worms	5	32	5	7.62±3.64	21.88±15.78	9.00±1.30
UNSW	AA-noAttMem	Fuzzers	5	3750	5	67.67±4.11	9.11±0.37	16.06±0.67
UNSW	AA-noAttMem	Worms	5	32	5	3.73±0.57	64.38±8.44	7.06±1.06
UNSW	Replay-BS (post hoc)	Fuzzers	5	3750	5	68.07±5.79	12.20±3.50	20.45±4.99
UNSW	Replay-BS (post hoc)	Worms	5	32	1	100.00	0.62±1.40	1.21±2.71
CIC17	AA-CIL	Botnet	5	370	5	6.75±0.96	95.41±1.74	12.59±1.66
CIC17	AA-CIL	Infiltration	5	6	5	2.17±1.25	66.67±0.00	4.16±2.34
CIC17	TS-mem	Botnet	5	370	5	0.72±0.29	5.73±2.00	1.25±0.48
CIC17	TS-mem	Infiltration	5	6	5	0.22±0.08	63.33±7.45	0.44±0.15
CIC17	AA-noAttMem	Botnet	5	370	5	6.50±1.01	96.05±2.38	12.15±1.74
CIC17	AA-noAttMem	Infiltration	5	6	5	3.36±2.00	66.67±0.00	6.31±3.53
CIC17	Replay-BS (post hoc)	Botnet	4	370	4	30.35±7.48	60.68±0.84	40.03±6.60
CIC17	Replay-BS (post hoc)	Infiltration	4	6	4	9.21±6.83	41.67±9.62	14.17±9.42

Precision uses defined seeds only; n_P is defined_n and n is the full completed seed count. Original null precision values remain null in the generated JSON. With no defined value the mean is –; with fewer than two the sample SD is undefined. No seed is removed from other defined endpoints. Complete per-class confusion matrices accompany the saved result JSON.

Table 56: Remaining registered contrasts. Differences are AA-CIL minus the named comparator. The harm tests use the less-than alternative; the CIC gate-refresh ablation uses the greater-than alternative.

Data	Endpoint	Comparator	Δ (pp)	p	Outcome
NSL	MF1 vs strongest	WACB	-1.365±5.922	0.31675	No detected harm
NSL	Guard vs Replay	Replay	14.534±7.440	0.99401	No detected harm
UNSW	MF1 vs strongest	AA-noAttMem	0.746±0.638	0.97037	No detected harm
UNSW	Guard vs Replay	Replay	33.232±0.851	1.00000	No detected harm
CIC17	MF1 vs strongest	AA-noAttMem	-0.455±1.756	0.29686	No detected harm
CIC17	Guard vs Replay	Replay	41.399±6.098	0.99995	No detected harm
CIC17	MF1 gate-refresh ablation	TS-mem	28.522±1.744	0.00000	PASS

Only complete five-seed groups determine these outcomes. No detected harm does not establish equivalence or noninferiority. Criterion 5 is descriptive completion of Joint-gap and cost reporting, not superiority over Joint.

Table 57: Per-seed recorded overhead across tasks (seconds, mean ± sample standard deviation). Separate post-training CPU analyses are shown without adding their nested subcomponents.

Data	Method	n	Selection	In-run calib.	In-run eval.	CPU curves	CPU ext. calib.
NSL	Replay	5	0.022±0.013	–	0.113±0.011	0.112±0.014	0.132±0.008
NSL	DER++	5	0.024±0.002	–	0.068±0.005	0.136±0.011	0.135±0.008
NSL	WACB	5	0.016±0.001	–	0.058±0.004	0.119±0.010	0.143±0.007
NSL	Freq	5	0.017±0.001	–	0.064±0.006	0.128±0.007	0.146±0.005
NSL	AA-CIL	5	0.082±0.003	0.025±0.010	0.081±0.013	0.058±0.004	–
NSL	TS-mem	5	0.027±0.001	0.020±0.003	0.079±0.006	0.063±0.017	–
NSL	AA-noAttMem	5	0.081±0.001	0.020±0.002	0.080±0.002	0.058±0.005	–
NSL	Replay-BS (post hoc)	5	0.083±0.004	0.325±0.016	0.058±0.004	0.110±0.028	–
NSL	Joint (offline)	5	0.012±0.001	–	0.059±0.003	0.149±0.019	0.154±0.007
UNSW	Replay	5	0.029±0.002	–	0.172±0.004	0.158±0.018	0.217±0.020
UNSW	DER++	5	0.036±0.003	–	0.085±0.008	0.197±0.017	0.236±0.022
UNSW	WACB	5	0.029±0.002	–	0.085±0.005	0.196±0.010	0.210±0.016
UNSW	Freq	5	0.027±0.001	–	0.080±0.004	0.184±0.009	0.197±0.015
UNSW	AA-CIL	5	0.074±0.006	0.022±0.001	0.118±0.007	0.061±0.004	–
UNSW	TS-mem	5	0.024±0.003	0.027±0.006	0.124±0.006	0.066±0.014	–
UNSW	AA-noAttMem	5	0.077±0.009	0.023±0.002	0.117±0.008	0.072±0.027	–
UNSW	Replay-BS (post hoc)	5	0.070±0.005	0.540±0.040	0.100±0.008	0.190±0.032	–
UNSW	Joint (offline)	5	0.022±0.002	–	0.087±0.003	0.228±0.006	0.213±0.019
CIC17	Replay	5	0.151±0.012	–	2.233±0.063	3.239±0.269	3.038±0.153
CIC17	DER++	5	0.187±0.021	–	1.246±0.099	3.674±0.131	3.257±0.176
CIC17	WACB	5	0.147±0.010	–	1.203±0.093	3.470±0.180	3.135±0.094
CIC17	Freq	5	0.149±0.006	–	1.204±0.068	3.623±0.061	3.170±0.173
CIC17	AA-CIL	5	0.291±0.018	0.374±0.126	1.919±0.061	1.216±0.081	–
CIC17	TS-mem	5	0.248±0.027	0.346±0.034	1.648±0.155	1.022±0.076	–
CIC17	AA-noAttMem	5	0.286±0.022	0.324±0.020	1.649±0.096	1.169±0.099	–
CIC17	Replay-BS (post hoc)	4	0.268±0.018	5.444±0.280	1.579±0.079	2.274±0.032	–
CIC17	Joint (offline)	5	0.032±0.001	–	1.198±0.023	3.672±0.124	3.216±0.149

Selection, in-run calibration and in-run evaluation use the recorded buffer-selection, calibration and evaluation totals respectively. A missing key is –, never an assumed zero. AA-family and Replay-BS CPU curves use the source-matched recorded cell-total seconds from the independent post-hoc curve ledger when available; missing or mismatched cell records remain –. For older baselines the separate CPU curve-analysis and extended-calibration elapsed totals are not embedded in the in-run columns; nested holdout/search/inference sub-timings must not be added again. Shared resume/render-attempt overhead remains in the independent curve ledger and is excluded here. These columns are not a single end-to-end walltime total and do not duplicate Replay+LA, which shares Replay weights and calibration.

References

[1] Suresh Kumar Amalapuram, Akash Tadwai, Reethu Vinta, et al. On handling class imbalance in continual learning based network intrusion detection systems. In *International Conference on AI-ML Systems*, 2021. doi: 10.1145/3486001.3486231. URL https://newslab.iith.ac.in/publications/on-handling-class-imbalance-in-continual-learning-based-network-intrusion-detection-systems/.

[2] Suresh Kumar Amalapuram, Bheemarjuna Reddy Tamma, and Sumohana S. Channappayya. Augmented memory replay-based continual learning approaches for network intrusion detection. In *Advances in Neural Information Processing Systems*, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/3755a02b1035fbadd5f93a022170e46f-Paper-Conference.pdf.

[3] Xinchen Zhang, Running Zhao, Zhihan Jiang, Handi Chen, Yulong Ding, Edith C. H. Ngai, and Shuang-Hua Yang. Continual learning with strategic selection and forgetting for network intrusion detection. In *IEEE INFOCOM*, 2025. URL https://arxiv.org/abs/2412.16264.

[4] Anomaly-aware class-incremental learning (ACIL) framework for network intrusion detection systems. *International Journal of Machine Learning and Cybernetics*, 17:104, 2026. doi: 10.1007/s13042-025-02840-5. URL https://link.springer.com/article/10.1007/s13042-025-02840-5.

[5] Cerasuolo et al. Class incremental learning for network-agnostic intrusion detection systems. In *IEEE International Forum on Research and Technologies for Society and Industry*, pages 17–22, 2025. doi: 10.1109/RTSI64020.2025.11212335.

[6] Suresh Kumar Amalapuram, Bheemarjuna Reddy Tamma, and Sumohana S. Channappayya. SPIDER: A semi-supervised continual learning-based network intrusion detection system. In *IEEE INFOCOM*, 2024. doi: 10.1109/INFOCOM52122.2024.10621428. URL https://newslab.iith.ac.in/media/publications/pdfs/SPIDER-Suresh.pdf.

[7] Sean Fuhrman, Onat Gungor, and Tajana Rosing. CND-IDS: Continual novelty detection for intrusion detection systems. In *ACM/IEEE Design Automation Conference*, 2025. URL https://arxiv.org/abs/2502.14094.

[8] Cerasuolo, Bovenzi, Ciuonzo, and Pescapè. Attack-adaptive network intrusion detection systems for IoT networks through class incremental learning. *Computer Networks*, 263:111228, 2025. doi: 10.1016/j.comnet.2025.111228.

[9] Cerasuolo, Bovenzi, Ciuonzo, and Pescapè. Adaptable, incremental, and explainable network intrusion detection systems for internet of things. *Engineering Applications of Artificial Intelligence*, 144: 110143, 2025. doi: 10.1016/j.engappai.2025.110143.

[10] Wang and Cao. A two-stage class incremental learning approach for network intrusion detection. In *IEEE Global Communications Conference*, pages 2353–2358, 2024. doi: 10.1109/GLOBECOM52923.2024.10901809. Official abstract and author publication record consulted; full text unavailable.

[11] Azizi Ariffin, Afif Haris, Faiz Zaki, Hazim Hanif, and Nor Badrul Anuar. Adaptive intrusion detection system using transformer-based neural networks and continual learning approach with adversarial investigation. *arXiv preprint arXiv:2608.04602*, 2026. URL https://arxiv.org/abs/2608.04602.

[12] Wu et al. Large scale incremental learning, 2019. URL https://arxiv.org/abs/1905.13260.

[13] Zhao et al. Maintaining discrimination and fairness in class incremental learning, 2020. URL https://arxiv.org/abs/1911.07053.

[14] Ahn et al. SS-IL: Separated softmax for incremental learning, 2021. URL https://arxiv.org/abs/2003.13947.

[15] Castro et al. End-to-end incremental learning, 2018. URL https://arxiv.org/abs/1807.09536.

[16] Menon et al. Long-tail learning via logit adjustment, 2021. URL https://arxiv.org/abs/2007.07314.

[17] Sylvestre-Alvise Rebuffi, Alexander Kolesnikov, Georg Sperl, and Christoph H. Lampert. iCaRL: Incremental classifier and representation learning. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 2001–2010, 2017.

[18] Pietro Buzzega, Matteo Boschini, Angelo Porrello, Davide Abati, and Simone Calderara. Dark experience for general continual learning: A strong, simple baseline. In *Advances in Neural Information Processing Systems*, volume 33, pages 15920–15930, 2020.

[19] Mahbod Tavallaee, Ebrahim Bagheri, Wei Lu, and Ali A. Ghorbani. A detailed analysis of the KDD CUP 99 data set. In *IEEE Symposium on Computational Intelligence for Security and Defense Applications*, pages 1–6, 2009.

[20] Nour Moustafa and Jill Slay. UNSW-NB15: A comprehensive data set for network intrusion detection systems. In *Military Communications and Information Systems Conference*, pages 1–6, 2015.

[21] Iman Sharafaldin, Arash Habibi Lashkari, and Ali A. Ghorbani. Toward generating a new intrusion detection dataset and intrusion traffic characterization. In *International Conference on Information Systems Security and Privacy*, pages 108–116, 2018.

[22] Stefan Axelsson. The base-rate fallacy and the difficulty of intrusion detection. *ACM Transactions on Information and System Security*, 3(3):186–205, 2000. doi: 10.1145/357830.357849.

[23] Robin Sommer and Vern Paxson. Outside the closed world: On using machine learning for network intrusion detection. In *2010 IEEE Symposium on Security and Privacy*, pages 305–316, 2010. doi: 10.1109/SP.2010.25.

[24] Daniel Arp, Erwin Quiring, Feargus Pendlebury, Alexander Warnecke, Fabio Pierazzi, Christian Wressnegger, Lorenzo Cavallaro, and Konrad Rieck. Dos and don'ts of machine learning in computer security. In *31st USENIX Security Symposium*, 2022. URL `https://arxiv.org/abs/2010.09470`.

[25] Hyrum S. Anderson and Phil Roth. EMBER: An open dataset for training static PE malware machine learning models. *arXiv preprint arXiv:1804.04637*, 2018. URL `https://arxiv.org/abs/1804.04637`.

[26] Yisroel Mirsky, Tomer Doitshman, Yuval Elovici, and Asaf Shabtai. Kitsune: An ensemble of autoencoders for online network intrusion detection. In *Network and Distributed System Security Symposium*, 2018. doi: 10.14722/ndss.2018.23204.

[27] Xin Tong, Yang Feng, and Jingyi Jessica Li. Neyman-pearson classification algorithms and np receiver operating characteristics. *Science Advances*, 4(2):eaao1659, 2018. doi: 10.1126/sciadv.aao1659.